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Hybrid forecasting of agricultural commodity prices: Integrating machine learning, time series, and stochastic simulation models

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ABSTRACT

This study examines a hybrid forecasting framework to evaluate the predictive performance of time series models (ARIMA, VAR), deep learning (LSTM), and stochastic simulations (GBM, FBM, BB) in forecasting agricultural commodity prices during global crises. Using daily data from 1985 to 2024, the analysis spans nine crisis periods, including the Global Financial Crisis, COVID-19, and the Russia–Ukraine conflict, and focuses on seven major agricultural commodities. Forecasting accuracy (MAPE, RMSE), risk (VaR), and return metrics are used to evaluate model performance. Results show that LSTM outperforms other models in capturing nonlinear dynamics during volatile episodes, whereas ARIMA provides stable results in shorter-term, low-volatility settings. GBM offers the best balance of forecast precision and risk-adjusted returns among stochastic models. In contrast, FBM captures memory effects but produces higher volatility. The findings highlight the importance of adaptive, context-specific forecasting models to enhance policy responses in food security, trade resilience, and agricultural risk management.

1. Introduction

Recent volatility in agricultural commodity prices plays a crucial role in global food security, economic stability, and the livelihoods of millions, particularly in developing regions. Accurate price forecasting is essential for farmers, policymakers, and market participants to anticipate fluctuations, optimize resource allocation, and develop strategies to mitigate risks associated with market instability. In response to these challenges, machine learning algorithms have gained significant attention in crop yield forecasting due to their ability to capture complex nonlinear relationships between predictor variables and target outcomes (Kang et al., 2020; Shahhosseini et al., 2021). However, traditional machine learning models often fall short of fully capturing the stochastic nature of agricultural commodity prices, which are shaped by diverse factors such as weather patterns, market dynamics, and geopolitical events.

A recent study by Ly et al. (2021) applies recurrent neural network (RNN) methods, specifically Long Short-Term Memory (LSTM) models, to forecast cotton and oil prices, comparing their performance with traditional methods like Autoregressive Integrated Moving Average (ARIMA). While LSTM models fit the data well, they did not consistently outperform classical approaches. However, combining forecasts from both methods significantly improves accuracy for cotton prices, reducing RMSE by 0.21 % and 21.49 % compared to ARIMA and LSTM, respectively. For oil prices, forecast averaging shows no improvement. The study highlighted the potential of forecast averaging and recommends extending the analysis to a broader range of commodities.

Historically, agricultural commodity prices have experienced significant fluctuations, driven by factors such as climatic events, geopolitical tensions, and global supply chain disruptions. One notable example is the 2007–2008 global food crisis during which a ban on agricultural exports was imposed. The food price crisis underscored the susceptibility of global agricultural markets to extreme price volatility, raising concerns about a potential shift to a new grain market regime marked by sustained price fluctuations. During this crisis, the prices of essential staples, including wheat, rice, and corn, surged significantly, driven by a confluence of factors such as rising oil prices, adverse weather conditions, and increased biofuel production (Headey & Fan, 2008; Trostle, 2008). For instance, Headey and Fan (2008) reported that global wheat prices more than doubled, while rice prices tripled within a few months, exacerbating food insecurity in low-income nations heavily reliant on these commodities for sustenance. The Asian Financial Crisis

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of 1997 had far-reaching impacts on emerging markets, especially in Southeast Asia. The crisis led to a sharp decline in the demand for agricultural exports due to weakened purchasing power and economic contraction in affected countries. Specifically, Warr (2002) found that Thailand's agricultural sector experienced a sheer drop in export revenues, particularly in rice, as regional demand declined.

The Global Financial Crisis (GFC) of 2008–2009 significantly disrupted commodity markets, leading to price surges across energy, metals, and agricultural sectors. In the agricultural commodity market, prices of essential staples like wheat, corn, and rice exhibited heightened volatility, driven by a confluence of collapsing global demand, restricted access to credit, and increased speculative trading (Tadesse et al., 2014). Subsequent studies illustrate that the post-GFC period saw tighter correlations between agricultural and energy markets, amplifying price shocks due to the rising use of biofuels (Baumeister & Kilian, 2014; Nazlioglu & Soytas, 2012). Additionally, fluctuations were further exacerbated by climate shocks, as demonstrated during the 2010 Russian wheat export ban, which caused global price spikes. Recent studies (Gardebroek et al., 2016; Kamal et al., 2021; Zhang & Broadstock, 2020) put forward that market integration and financialization have made agricultural markets more sensitive to global economic shocks, requiring robust policy interventions to mitigate risks (Glauber et al., 2020).

More recently, the COVID-19 pandemic has introduced significant uncertainty into agricultural commodity markets, disrupting traditional supply and demand dynamics and increasing market volatility. Supply chain interruptions, such as port closures and logistical delays, significantly affected the global movement of commodities like wheat, corn, and soybeans, as highlighted by Laborde et al. (2020). Labor shortages caused by lockdowns and mobility restrictions further hampered agricultural production, particularly in labor-intensive sectors, exacerbating price instability (OECD, 2021). Furthermore, shifts in consumer behavior, including increased online food purchases and changes in dietary patterns, contributed to price fluctuations in specific markets, as mentioned by Espitia et al. (2022). Financial speculation in agricultural futures markets further amplified price volatility during the pandemic. Ullah et al. (2023) conclude that increased investor activity in commodity trading, driven by economic uncertainty, led to heightened volatility in agricultural markets.

Overall, the persistent volatility of agricultural commodity prices underscores their sensitivity to macroeconomic fluctuations and geopolitical events. As global food systems become increasingly interconnected and exposed to systemic shocks from financial crises and pandemics to armed conflicts accurate price forecasting has become essential for ensuring food security, stabilizing markets, and guiding policy responses. Effective forecasting tools are exceptionally vital for stakeholders such as producers, investors, and policymakers who must navigate short-term disruptions and long-term structural changes in agricultural markets.

To address these challenges, this study advances the literature on agricultural commodity price forecasting by conducting a comprehensive comparative analysis of both data-driven and stochastic modeling techniques. Specifically, it evaluates the predictive performance of classical time series models (ARIMA, VAR), deep learning approaches (LSTM), and stochastic processes (Geometric Brownian Motion (GBM), Fractional Brownian Motion (FBM), and Brownian Bridge (BB)). Unlike prior studies that focus narrowly on single methods or limited time spans, this study leverages daily price data for seven major agricultural commodities, cocoa, coffee, corn, cotton, soybean, sugar, and wheat, over a broad timeline spanning from July 19, 1985, to April 19, 2024.

A key contribution of this study lies in its multi-crisis analytical framework. By segmenting the data across nine distinct global crisis episodes ranging from the Farm Crisis and the Asian Financial Crisis to the Global Financial Crisis, COVID-19 pandemic, and Russia–Ukraine conflict, the study scrutinizes the performance of LSTM, ARIMA, and VAR models relative to traditional stochastic approaches under varying levels of market volatility, notably during periods of heightened uncertainty. This regime-based approach enables a nuanced understanding of model robustness and adaptability under dynamic economic and geopolitical conditions.

The findings shed light not only on the strengths of LSTM models in capturing non-linear and long-memory dynamics, but also on the relative resilience of simpler models like ARIMA during short-term or moderately volatile phases. LSTM models generally offer the highest return potential, especially during highly volatile regimes, though often with elevated error rates. In contrast, ARIMA consistently delivers low forecast errors, particularly in stable or early-stage crises, albeit with minimal return insights. GBM emerges as the most robust stochastic model, balancing accuracy and risk-adjusted returns during turbulent episodes, while FBM captures speculative dynamics with higher volatility. The BB model underperforms across all scenarios.

The remainder of the paper is organized as follows: Section 2 provides a review of relevant literature. Section 3 outlines the data and methodology. Section 4 presents our empirical results. Finally, section 5 discusses key conclusions and policy implications.

2. Literature review

Forecasting agricultural commodity prices is a critical component of risk management for stakeholders such as farmers, traders, and policymakers. Accurate predictions of price movements enable informed decision-making related to production planning, market participation, and policy formulation. The agricultural sector, which once suffered from extensive price support and stabilization mechanisms, now faces heightened exposure to market-driven volatility. As noted by the OECD (2009, p. 168), agriculture is subject to a broad array of risks, varying from production and market fluctuations to institutional and financial uncertainties. In response, researchers have increasingly explored advanced forecasting methodologies beyond traditional econometric approaches. Recent developments in stochastic modeling and machine learning, especially LSTM networks, offer promising alternatives for capturing complex price dynamics and improving forecast precision. This literature review offers a comparative synthesis of classical time series models, stochastic simulation techniques, and deep learning approaches, focusing on their respective strengths and limitations in forecasting agricultural commodity prices under varying economic conditions.

Several studies have identified the factors driving recent fluctuations in agricultural commodity prices. One of the pioneering studies is Deaton and Laroque (1992), who examined the behavior of commodity prices using a rational expectations storage model. Their study illustrated the significant role of supply shocks and inventory levels in driving price volatility, demonstrating that commodity prices often exhibit mean-reverting behavior with occasional sharp spikes. Similarly, Cashin and McDermott (2002) provided a comprehensive analysis of the persistence and volatility of commodity prices over the long term. They found that commodity prices tend to follow cyclical patterns influenced by both short-term market dynamics and longer-term structural changes.

Moreover, Abbott and Borot de Battisti (2009) and Gilbert (2010) further extended this line of research by investigating the factors behind the 2007–2008 global food price crisis. Additionally, they highlighted the role of speculative activities in financial markets, which amplified price movements during the crisis. Later, Gilbert and Morgan (2010) revealed the influence of macroeconomic factors on agricultural commodity prices, focusing on the impact of oil price fluctuations and currency exchange rates. They demonstrated that changes in crude oil prices significantly affect agricultural production costs and commodity prices.

Traditional econometric models, including ARIMA and Generalized Autoregressive Conditional Heteroskedasticity (GARCH), have been widely used due to their ability to model time series data and volatility

in commodity prices. Nevertheless, these models often struggle to capture the nonlinear patterns and intricate interactions present in agricultural markets, in particular during periods of high volatility or structural breaks. One of the studies by Nazlioglu et al. (2013) examines the dynamic relationship between agricultural commodity prices and macroeconomic factors using a vector error correction model (VECM). Their findings highlight the substantial impact of global oil prices and fluctuations in exchange rates on agricultural commodity markets. This study underscores the limitations of traditional econometric models in capturing these external influences, particularly during periods of economic uncertainty. Tansuchat et al. (2009) analyze the volatility of 16 agricultural commodity futures returns across various markets, including corn, wheat, soybeans, cocoa, and coffee, using fractional GARCH models. Their findings indicate significant long-term dependence in most commodity futures, with fractional models outperforming traditional GARCH and EGARCH models in capturing volatility dynamics.

Recently, Balcilar et al. (2021) examined the impact of COVID-19 on agricultural commodity prices and volatility from 2016 to 2022 using a news-based sentiment index. A nonparametric Granger causality in quantiles test reveals that negative COVID-19 sentiment significantly affects prices and drives volatility, particularly in higher quantiles. The findings demonstrate that COVID-19 caused both price crashes and increased market risk. Dinku and Worku (2022) investigate the volatility of agricultural commodity prices in Ethiopia from 2010 to 2020 using EGARCH and TGARCH models. The EGARCH model best fit prices for maize, and others, while TGARCH was better for sorghum, barley, and beans. The findings indicated time-varying volatility and a leverage effect, where positive and negative news affected prices differently. Similarly, Abid et al. (2023) analyze the impact of Geopolitical Risk (GPR) shocks on five commodity types (energy, precious metals, agriculture, industrial metals, and livestock) using a Markov-Switching model. The results illustrate that all commodities are affected by GPR shocks, with the energy market being the most reactive and livestock the least. Lastly, Dutta et al. (2024) investigate the impact of various uncertainty measures on the volatility of agricultural futures markets and compare news-based uncertainty with market-based measures. Using LASSO and HAR methods, the findings show that market-based uncertainty measures offer better volatility forecasts, with LASSO outperforming HAR in out-of-sample predictions.

Several studies have shed light on the use of ML algorithms for crop yield forecasting and agricultural commodity price prediction, demonstrating their efficacy in handling complex and non-linear relationships within agricultural data. Kamilaris and Prenafeta-Boldú (2018) illustrate the use of deep learning models, such as neural networks, for crop yield forecasting, integrating data like satellite imagery, weather forecasts, and soil conditions. They find that deep learning outperformed traditional regression models due to its ability to capture complex relationships between environmental and crop data. Similarly, Paul et al. (2022) explore the use of ML algorithms for forecasting agricultural prices in India and depict that ML methods, particularly GRNN, outperform ARIMA in forecasting accuracy. Additionally, Liu et al. (2022) investigate a hybrid model combining CEEMDAN decomposition and a CNN-GRU-Attention network to predict China's soybean spot prices. The model outperformed others in prediction accuracy, reducing RMSE, MAPE, and MAE by significant margins.

Furthermore, Göncü et al. (2024) compared machine learning algorithms, including logistic regression, neural networks, SVM, XGBoost, ExtraTrees, random forests, and CNN, for predicting daily spot oil price trends over 1–8 days. Using technical indicators, financial data, and volatility measures, the CNN model surpasses others, likely due to its ability to capture visual patterns in price movements, similar to how traders identify trends. Another notable study by Zhang and Tang (2024) applies a VMD-SGMD-LSTM model for forecasting agricultural futures prices. The model decomposes data using VMD and processes it with SGMD, uses LSTM to predict and combine the results. Their empirical tests on wheat, corn, and sugar futures present that this model outperformed others in forecasting accuracy and robustness.

Sari et al. (2024) studied a novel agricultural commodity price forecasting model using an extreme learning machine (ELM) optimized with a genetic algorithm. Applied to predict the prices of eleven key commodities, the ELM model excels beyond other methods, such as LSTM with a genetic algorithm and ARIMA. Despite market fluctuations in 2022, the model provides accurate predictions, offering valuable insights for policymakers in managing agricultural policies and food security. Several other studies (Pandit et al., 2024; Patil et al., 2023; Paul et al., 2022; Rana et al., 2024) also employ ML techniques for the prediction of agricultural commodity prices in order to select the best alternative model.

More significantly, LSTM networks have gained prominence due to their ability to learn long-term dependencies in sequential data. Sherstinsky (2018) elucidates the fundamental principles of recurrent neural networks (RNNs) and LSTM architectures, emphasizing their applicability in time-series forecasting, which is pivotal in agricultural price predictions. Additionally, Zhang et al. (2018) developed an LSTM-based model primarily for predicting water table depth, demonstrating the model's versatility in handling agricultural-related metrics. Other studies by Lindemann et al. (2021) and Ghimire et al. (2021) further reinforced LSTM's strengths in time series forecasting, indicating its robustness across various environmental variables. Additionally, the integration of attention mechanisms in LSTM networks, as explored by Zhou et al. (2016) and Qiu et al. (2020), enhanced the model's performance by allowing it to focus on relevant parts of the input data, thus improving prediction accuracy.

Over time, Mahto et al. (2021) applied artificial neural networks (ANN) with the traditional ARIMA model for short-term price forecasting of sunflower seed and soybean seed. The results demonstrate that ANN outperformed ARIMA, with better performance based on mean absolute percentage error (MAPE) and root mean square percentage error (RMSPE), making it a more effective tool for agricultural price predictions. Another study by Ray et al. (2022) explores machine learning, particularly deep learning, as an alternative to traditional statistical models like ARIMA and GARCH for forecasting complex time series. The results illustrate that the ARIMA-LSTM model outperforms conventional models in forecasting accuracy, with improvements of 33–71 % for RMSE and 33–77 % for MAPE.

Moreover, Xia (2024) compares ARIMA and LSTM models for predicting agricultural commodity prices, focusing on greens and lotus roots. The results present that ARIMA outperforms LSTM in accuracy, with lower error metrics, especially for lotus roots. Despite LSTM's complexity, ARIMA remains a more reliable and efficient tool for agricultural price forecasting. Rana et al. (2024) forecast agricultural commodity prices in Pakistan using ARIMA, Random Forest, and LSTM models on historical spinach price data. LSTM outperformed the other models, with the highest R^2 and lowest Mean Absolute Error (MAE), providing valuable insights for farmers and traders. More significantly, various studies (Hernandez et al., 2019; Ihsan et al., 2017; Oktoviany et al., 2021; Prasad et al., 2019; Sun et al., 2023; von Bloh et al., 2024) conclude that the LSTM model outperforms conventional statistical methods, such as ARIMA and GARCH, in forecasting agricultural commodity prices. LSTM's ability to capture complex, non-linear patterns in time series data makes it particularly effective for predicting price trends, even in volatile markets.

In addition to machine learning, stochastic models have also been employed to capture the inherent uncertainty in agricultural commodity prices. The LSTM model's predictive capabilities are often compared with other stochastic models. GBM is widely used in financial modeling, particularly for stock prices, due to its properties of continuous price movements and its incorporation of randomness. Chen et al. (2021) highlighted the application of machine learning in mean-variance portfolio optimization, indicating that GBM can effectively model price fluctuations in commodities. Pindyck (1999) and Schwartz (1997)

are among the early researchers who applied GBM to model commodity prices, oil in particular.

Furthermore, Farida Agustini et al., (2018) investigate the GBM model for predicting stock prices by first determining the expected cost. They indicate that the GBM model provides highly accurate predictions, with a forecast MAPE value of 20 % or lower. Z. Muda and Nur Iqmal Ibrahim, 2023 compare three models as GBM, Geometric Fractional Brownian Motion (GFBM), and Merton Jump-Diffusion (MJD) to forecast Malaysia's rubber prices. The MJD model outperforms the others, attaining the highest forecast accuracy, as measured by the lowest MAPE.

While previous studies have demonstrated the utility of both machine learning and stochastic models in commodity price forecasting, particularly in financial and energy markets, their comparative performance in agricultural commodity markets remains underexplored, especially in the context of recurrent global shocks. Moreover, numerous prior works are limited by narrow temporal scopes or focus on a single commodity, thereby restricting the generalizability of their findings. This study fills this gap by taking these limitations by leveraging a comprehensive dataset spanning nearly four decades (beginning on July 19, 1985, and ending on April 19, 2024), encompassing nine distinct global crisis periods, and analyzing seven major agricultural commodities individually and comparatively. By merging classical time series methods (ARIMA, VAR), deep learning (LSTM), and stochastic simulation models (GBM, FBM, BB), this study offers one of the most extensive and systematic evaluations to date. The multi-period, multi-commodity, and multi-model framework provides nuanced insights into model performance under varying structural regimes, enhancing the understanding of forecasting reliability amid economic and geopolitical turbulence.

3. Methodology and data

3.1. Methodology

3.1.1. Geometric Brownian Motion

Geometric Brownian Motion (GBM) is a stochastic process commonly used to describe the time dynamics of prices of financial assets. The price of an asset, $S(t)$, is defined by the following differential Equation (1):

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t) \tag{1}$$

In Equation (1), $S(t)$ is the asset price at time t, μ is the expected rate of return (drift), σ, is the volatility of the asset, and $W(t)$ is the standard Brownian motion. Based on Algorithm 1, each price path is calculated by Equation (2).

$$S(t+\Delta t) = S(t) \times exp\left(\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} \times Z_t\right) \tag{2}$$

In Equation (2), Δt, denotes the time step and $Z_t : N(0,1)$ denotes the normal distribution. Upon simulating n Brownian motions, the average price path $AGBM(t)$ at each time step t can be calculated by Equation (3).

$$AGBM(t) = \frac{1}{n}\sum_{i=1}^{n} S_i(t) \tag{3}$$

This equation gives the average value at each time step over n simulation paths.

Algorithm 1. Geometric Brownian Motion

```
procedure Simulate_Geometric_Brownian_Bridge(S0, μ, σ, T, dt, N)
Time_Points← Linspace(0, T, T/dt+1)
GBM_Paths← Zeros(N, T/dt+1)
GBM_Paths[:,0]← S0
Repeat
  for i←1 to N do
    for t←1 to T/dt do
      Z ← Random_Normal(0,1)
      GBM_Paths[i,t] ← GBM_Paths[i,t-1] × exp((μ-0.5 × σ^2) ×dt + σ × √dt × Z
    end for
  end for
return GBM_Paths
end procedure
```

3.1.2. Fractional Brownian Motion

Fractional Brownian Motion (FBM) is a generalization of standard Brownian motion and is characterized by the Hurst parameter H. The Hurst parameter takes a value between $0<H<1$. This parameter controls the dependence in the process. When $H = 0.5$, FBM is equivalent to standard Brownian motion. $H > 0.5$ means long-term positive dependence and $H < 0.5$ means long-term negative dependence. The FBM process is modeled as in Equation (4):

$$B_H(t) = \int_0^t (t-s)^{H-\frac{1}{2}} dW(s) \tag{4}$$

where, $B_H(t)$ is the fractional Brownian motion defined by the Hurst parameter H and $dW(s)$ is the differential of the standard Brownian motion. Based on Algorithm 2, each price path is calculated by Equation (5).

$$FBMPaths_i(t) = S_0 + \sum_{j=1}^{t} N\left(0, \sqrt{dt}\right) \times \text{Cholesky}(CovMatrix) \tag{5}$$

If multiple FBM simulations are performed, the average price path at each time step t can be calculated by Equation (6):

$$AFBM(t) = \frac{1}{n}\sum_{i=1}^{n} FBMPaths_i(t) \tag{6}$$

This equation gives the average value at each time step over n simulation paths.

Algorithm 2. Fractional Brownian Motion

```
procedure Simulate_ Fractional _Brownian_Bridge(H, T, dt, N)
Time_Points← Linspace(0, T, T/dt+1)
FBM_Paths←Zeros(N, T/dt+1)
Repeat
  for i←1 to N do
    generate FBM_ Increments based on Hurst parameter H
    compute FBM_Paths[i,:] ←Cumsum(FBM_ Increments)
    end for
  end for
return FBM_Paths
end procedure
```

3.1.3. Brownian Bridge

A Brownian Bridge is a version of standard Brownian motion with specific starting and ending points. In the Brownian Bridge process, the value of the Brownian motion expressed as $B(t)$ $B(t)$ at time $t = T$ is constrained by $B(T)$. The Brownian Bridge can be modeled as follows:

$$B(t) = W(t) - \frac{t}{T}W(t) \tag{7}$$

In Equation (7), B(t) is the BB at time t, W(t) is the standard Brownian motion at time t, T is the final time point. This equation ensures that the process B(t) returns to zero at time t = T. Since W(t) is the standard Brownian motion, W(t) is considered as a normally distributed random variable. In Algorithm 3, dt is defined as the time step and the total number of steps for each simulation is $N = t/dt+1$. To simulate Brownian motion at each time step t during time period t, the normal distribution N $(0, \sqrt{dt})$ is used. Based on Algorithm 3, each price path is calculated by

Equation (8).

$$BridgePaths_i(t) = S_0 + \sum_{j=1}^{t} N\left(0, \sqrt{dt}\right) - \frac{t}{T}\sum_{j=1}^{t} N\left(0, \sqrt{dt}\right) \quad (8)$$

This formula calculates the value of the Brownian Bridge at each time step t for the i-th simulation. Upon performing n Brownian Bridge simulations, the average price path $ABB(t)$ at each time step t can be calculated by Equation (9).

$$ABB(t) = \frac{1}{n}\sum_{i=1}^{n} BridgePaths_i(t) \quad (9)$$

This equation gives the average value at each time step over n simulation paths.

Algorithm 3. Brownian Bridge

```
procedure Simulate_Brownian_Bridge(S0, T, dt, N)
Time_Points← Linspace(0, T, T/dt+1)
Bridge_Paths←Zeros(N, T/dt+1)
Repeat
  for i←1 to N do
      Brownian_motion←Cumsum(Random_Normal(0, √dt, T/dt + 1))
    for each t in Time_Points do
      Bridge_Paths[i,t] ← S0 + Brownian_motion[t]- (Time_Points[t]/T)× Brownian_motion[-1]
    end for
  end for
return Bridge_Paths
end procedure
```

3.1.4. Long Short-Term Memory (LSTM)

A specific type of Recurrent Neural Networks (RNNs) called Long Short-Term Memory (LSTM) networks is developed to learn long-term dependencies in data sequences (Sherstinsky, 2018). The fact that LSTMs can keep and update a cell state for an extended period gives them a particular advantage in regression problems using time series data. The critical elements of LSTM are the cell state and the gate mechanism. Cell states can be used to pass the information and prevent the influence of short-term memory. The three gate structures in LSTMs are the input gate, the forgetting gate, and the output gate, which serve different purposes.

The input gate modifies the cell state. The sigmoid function takes the previous hidden state and the current input as inputs and changes the value between 0 and 1 to decide what should be updated.

$$i_t = \sigma(\omega_i[h_{t-1}, x_t] + b_i) \quad (10)$$

The LSTM uses the forgetting gate to assist it choose which data to discard.

$$f_t = \sigma\left(\omega_f[h_{t-1}, x_t] + b_f\right) \quad (11)$$

Next, the cell state is calculated. This process begins by multiplying the forgetting gate vector with the previous cell state. The new cell state has to be multiplied by a factor close to 1 for the information to be retained.

The newly obtained cell state passes through the tanh activation function, while the sigmoid function takes the current input and the previous hidden state. Then, the knowledge that the hidden state should contain is decided by the product of the outputs obtained from the tanh activation function and the sigmoid activation function. The output of the current cell is its hidden state, and the next time step uses both the updated cell state and the new hidden state.

$$\widetilde{C}_t = tanh(w_c[h_{t-1}, x_t] + b_c) \quad (12)$$

$$C_t = f_t^* C_{t-1} + i_t^* \widetilde{C}_t \quad (13)$$

$$o_t = \sigma(\omega_0[h_{t-1}, x_t] + b_0) \quad (14)$$

$$h_t = o_t^* \tanh\left(C_t\right) \quad (15)$$

where i_t, f_t, o_t, $\widetilde{C}_t$, and C_t represent the vector of the memory cell, the input gate, the forgetting gate, the output gate, and the candidate value added to the new cell state. ω_i, ω_f, ω_0, ω_c represent the weight matrix. For the bias vectors, use b_i, b_f, b_0, and b_c. The sigmoid function is activated by σ. The resulting matrix is h_t. The output vector is $(y_1, y_2, \cdots, y_t)$, and the error function is written as in Equation (16). The goal is to minimize the sum of squares of the difference between the output H_t and the target y_t. The pertinent parameters for feature extraction and prediction are updated by a back propagation method.

$$E_r = Min\left(\sum_{t=1}^{M}\left(H_t - y_t\right)^2\right) \quad (16)$$

3.1.5. Autoregressive Integrated Moving Average (ARIMA)

To establish a benchmark for time series forecasting, we employed the Autoregressive Integrated Moving Average (ARIMA) model, which has been widely adopted in the literature for univariate time series modeling. The ARIMA model captures linear dependencies in stationary series by combining autoregressive (AR), differencing (I), and moving average (MA) components, typically denoted as ARIMA (p, d, q).

The order of differencing (d) was determined based on stationarity tests using the Augmented Dickey-Fuller (ADF) test. Upon achieving stationarity, we chose optimal lag orders p and q by minimizing the Akaike Information Criterion (AIC) over a predefined grid of model configurations. The model was then fitted using maximum likelihood estimation, and one-step-ahead recursive forecasts were generated for each out-of-sample period.

This model provides a robust baseline against which we compare more complex, nonlinear, and multivariate methods.

3.1.6. Vector Autoregression (VAR)

To capture the dynamic interdependence within time-lagged structures of the time series, we also utilized the Vector Autoregression (VAR) model. Unlike ARIMA, which is limited to univariate series, VAR allows for modeling multiple time series simultaneously, treating each variable as a linear function of its own past values and those of other variables in the system. Since our forecasting framework focuses primarily on price series, we constructed a VAR model using the lagged level of the price variable and its first lag as explanatory variables. The lag order was selected based on the Schwarz Information Criterion (SIC), and model stability was verified by ensuring all eigenvalues lie within the unit circle.

Although we consider only lagged price levels in our implementation (i.e., a bivariate structure with one lag), the VAR framework is extendable to incorporate additional exogenous variables or multivariate price systems across related commodities.

Table 1 summarizes the key configuration parameters used for the ARIMA, VAR, and LSTM models implemented in this study. The ARIMA model parameters were set to (5,1,0), where the differencing order d = 1 was determined using the Augmented Dickey-Fuller (ADF) test to ensure stationarity, whereas the autoregressive and moving average terms were fixed to ensure comparability across different periods. The VAR model was implemented with a single lag (VAR(1)), based on a parsimonious approach due to the bivariate structure of the model, which only included the original price and its first lag.

The LSTM model was constructed with a simple architecture to reduce overfitting and computational overhead. A look-back window of 10 time steps was selected to capture recent temporal dependencies, and a single-layer LSTM with 32 hidden units was used. The learning rate and batch size were set to standard using default values (0.001 and 32, respectively), and the model was trained over 20 epochs. These choices

Table 1
Model Configuration parameters.

Model	Parameter(s)	Value (s)	Selection Criteria
ARIMA	(p, d, q)	(5, 1, 0)	Selected based on ADF test for stationarity (d) and predefined structure (p = 5, q = 0) for comparability.
VAR	Lag order (p)	1	Fixed at 1 to avoid overfitting due to the limited bivariate structure.
LSTM	Look-back window	10	Manually chosen based on domain heuristics.
	Hidden size	32	Selected to balance capacity and computational efficiency.
	Number of LSTM layers	1	Simpler structure to avoid overfitting.
	Batch size	32	Standard mini-batch training configuration.
	Learning rate	0.001	Common default for Adam optimizer
	Epochs	20	Chosen as a trade-off between convergence and training time

reflect a balance between computational feasibility and forecast accuracy, particularly given the high volatility and nonlinearity present in commodity price data.

3.1.7. Evaluation metrics

In this study, two individual evaluation metrics, the mean absolute percentage error (MAPE) and root mean square error (RMSE), were applied in order to evaluate model performance. The RMSE gives the deviation between the actual value and predicted value, and it is given as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2} \tag{17}$$

The difference between the actual and anticipated values is expressed as a percentage using the MAPE metric, which is frequently used in time series prediction. The following is the definition of MAPE:

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y_i}}{y_i}\right| \times 100 \tag{18}$$

The variables n and $\widehat{y_i}$ represent the number of data samples and the real and derived values of the i -th sample data, respectively, in Equations (17) and (18). Table 2 summarizes the ruling on any forecasting technique utilizing MAPE by Lawrence et al. (2006). RMSE is calculated by deducting each data sample's actual value from its anticipated value. The root operation is then carried out upon calculating the average of the square values. To calculate MAPE, the absolute difference between each actual and predicted value is divided by the actual value, and then the mean of these percentage errors is calculated. MAPE is particularly useful for interpretability, since it expresses the error as a percentage regardless of the scale of the predicted values.

3.1.8. Value at risk

Value at Risk (VaR_{90}) at the 90 % confidence level represents the maximum expected loss over the forecast horizon, based on the 10th percentile of the distribution of simulated returns.

$$VaR_{90} = -Percentile_{10}\left(\frac{S_T^{(i)} - S_0}{S_0}\right) \times 100 \tag{19}$$

Table 2
MAPE judgment accuracy of forecasting method.

Judgment of Accuracy	MAPE
Highly accurate	MAPE <10 %
Good accuracy	10 % ≤ MAPE <20 %
Reasonable	20 % ≤ MAPE <50 %
Inaccurate	MAPE ≥50 %

where, S_0, initial price at the start of simulation, $S_T^{(i)}$, terminal price of the i-th simulation path, $(S_T^{(i)}\text{-}S_0)/\ S_0$, return over the simulation period, The 10th percentile of the return distribution is taken and multiplied by −1 to express VaR as a positive loss figure.

3.1.8.1. Average Return. Average Return represents the arithmetic mean of the end-period returns across all simulated paths.

$$Average\ Return = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{S_T^{(i)} - S_0}{S_0}\right) \tag{20}$$

where, N, number of simulation paths, $S_T^{(i)}$, terminal price of the i-th simulation path, S_0, initial price.

3.1.9. Average Sharpe Ratio

The Sharpe ratio measures risk-adjusted return and is calculated for each simulation path as the ratio of average return to standard deviation of returns. The reported value is the mean Sharpe ratio across all simulation paths.

$$sharpe^{(i)} = \frac{\mu_i}{\sigma_i}, Average\ Sharpe = \frac{1}{N}\sum_{i=1}^{N} sharpe^{(i)} \tag{21}$$

where, μ_i, mean return of the i-th simulation path, σ_i, standard deviation of returns in the i-th path. The risk-free rate is assumed to be zero in this study.

To evaluate the risk and performance of the simulated price paths, we employed three key metrics: Value at Risk (VaR_{90}), Average Return, and Average Sharpe Ratio. VaR_{90} measures the potential downside risk under a 90 % confidence level. Average Return captures the overall profitability of the simulated paths, while the Average Sharpe Ratio measures the risk-adjusted performance by considering return volatility across each simulated scenario. In summary, this study uses MAPE, RMSE, VaR_{90}, Average Return, and Average Sharpe Ratio to evaluate forecasting accuracy and risk-adjusted performance.

As shown in Fig. 1, the proposed framework integrates both stochastic models (GBM, FBM, BB) and data-driven methods (LSTM, ARIMA, VAR) across nine global crisis periods and seven major agricultural commodities using historical daily data spanning from 1985 to 2024.

3.2. Data and description

This section outlines the dataset and provides a detailed description of the agricultural commodities analyzed. The study utilizes a high-frequency dataset comprising daily spot and futures prices for seven key agricultural commodities: wheat, corn, soybean, coffee, sugar, cocoa, and cotton. The data were sourced from reputable international sources, including the U.S. Department of Agriculture, the International Coffee Organization (ICO), the International Cocoa Organization (ICCO), and the International Sugar Organization (ISO). Summary information about the commodity series, including commodity codes, underlying spot price definitions, and data sources, is presented in Table 3.

To analyze the dynamic behavior of commodity prices, this study employs both stochastic models, including GBM, FBM, BB, and time-series and machine learning models, such as ARIMA, VAR, and LSTM neural networks. Forecasts are generated on a regime-based (periodic) basis, corresponding to nine distinct global crisis periods, such as financial meltdowns, geopolitical tensions, and pandemic shocks. The objective is to provide a comprehensive assessment of model performance under varying structural conditions by capturing both random walk properties and non-linear dependencies in commodity price movements.

The analysis in this study is structured around significant global

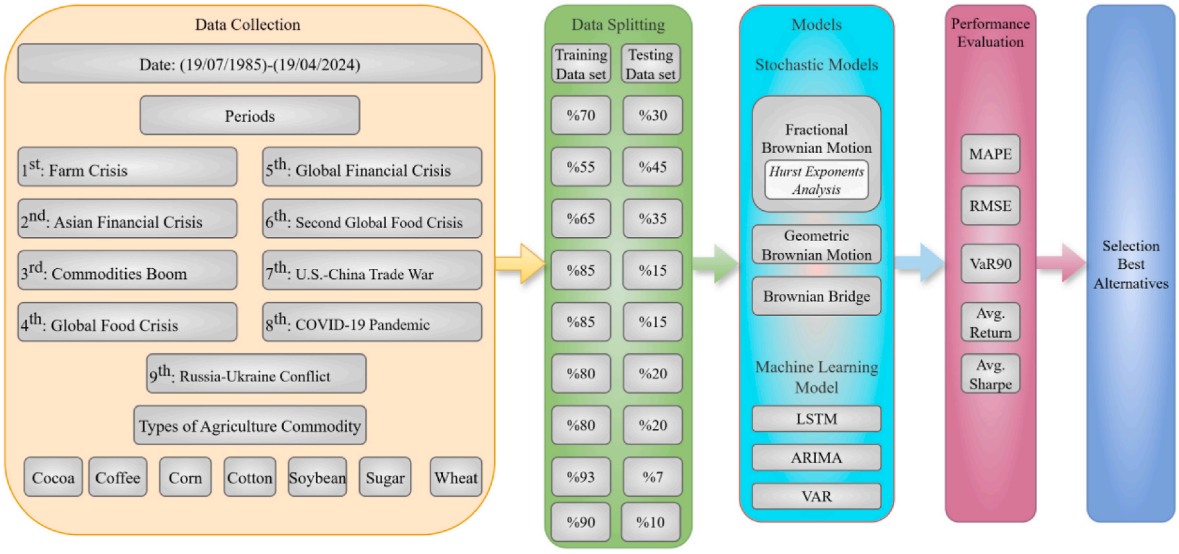


Fig. 1. The flow chart diagram process.

Table 3
Data information.

Commodity	Symbol	Underlying Spot Commodity	Source
Wheat	WHT	Wheat Number 2, Soft Red USD/Bushel	U.S. Department of Agriculture
Corn	CRN	Corn Number 2 Yellow Central Illinois USD/Bushel	U.S. Department of Agriculture
Soybean	SBN	Soyabeans, Number 1 Yellow USD/Bushel	U.S. Department of Agriculture
Coffee	COF	Coffee-International Coffee Organization (ICO) Brazilian Natural Average UC/Pound	International Coffee Organization (ICO)
Sugar	SGR	Raw Sugar-International Sugar Agreement (ISA) Daily Price UC/Pound	International Sugar Organization (ISO)
Cocoa	CCA	Cocoa-International Cocoa Organization(ICCO) Daily Price USA United States Dollar Per Metric Tonne	International Cocoa Organization (ICCO)
Cotton	CTN	Cotton, 1 1/16STR Low - Middling, Memphis USD/Pound	U.S. Department of Agriculture

crises and events that have shaped agricultural commodity markets over time. It begins with the Farm Crisis of the 1980s, which highlighted the vulnerability of agricultural sectors to debt-induced collapses and shifts in trade policies. The Asian Financial Crisis (1997–2000) follows, demonstrating how regional currency devaluations and economic contractions affected export-driven agricultural economies, causing price fluctuations across staples like rice and wheat.

Next, the Global Commodity Boom (2003–2008) is examined, a period of rising commodity prices driven by rapid industrialization in emerging economies such as China and India. This is followed by the Global Financial Crisis (2008–2009), during which collapsing financial markets led to declining demand and sharp volatility in agricultural prices. The Second Global Food Crisis (2010–2012), fueled by a combination of weather shocks and policy-driven export restrictions, further highlighted the fragility of global food systems.

The U.S.-China Trade War (2018–2020) introduced new uncertainties, as retaliatory tariffs disrupted global supply chains and shifted agricultural trade flows, particularly affecting soybean and corn markets. The COVID-19 Pandemic (2020–2022) presented unprecedented challenges, with lockdowns, labor shortages, and supply chain disruptions causing price volatility and demand shifts in essential commodities.

Finally, the ongoing Russia-Ukraine Conflict (2022–present) adds complexity to the analysis, as geopolitical tensions and disruptions in grain exports from two of the world's largest suppliers have led to global price surges, particularly in wheat and sunflower oil, exacerbating food insecurity in import-dependent regions. This comprehensive exploration of macroeconomic events provides valuable insights into how external shocks affect the dynamics of agricultural commodity prices.

4. Empirical findings

4.1. Hurst exponents analysis

The determination of the Hurst exponent is usually done using techniques such as the Rescaled Range (R/S) approach, the Variance-Time approach, or the detrended fluctuation analysis (DFA). In this paper, the Hurst exponent is estimated using R/S analysis, which is particularly appropriate for analyzing financial time series. The computed Hurst exponents reveal trends relative to major market events.

4.2. GBM, FBM, BB, LSTM, ARIMA, and VAR analysis

Upon the Hurst exponent analysis, which is a critical parameter in FBM analysis, GBM, FBM, BB, LSTM, ARIMA, and VAR analyses were examined. FBM generalizes standard Brownian Motion by allowing long-term dependence on the data. The Hurst exponent directly affects FBM behavior.

Several factors affect prediction accuracy, including model selection, parameter estimation, and the quality and representativeness of training data. We collected worldwide index data for wheat, corn, soybean, coffee, sugar, cocoa, and cotton over 10,111 days, from July 19, 1985, to April 19, 2024. We used the training set to determine the parameters needed for the simulation after dividing the data into training and test sets with different proportions for each period (Farm Crisis 70 % training, 30 % testing, Asian Financial Crisis 55 % training, 45 % testing, Commodities Boom 65 % training, 35 % testing, Global Food Crisis 85 % training, 15 % testing, Global Financial Crisis 85 % training, 15 % testing, Second Global Food Crisis 80 % training, 20 % testing, U.S.-China Trade War 88 % training, 20 % testing, COVID-19 Pandemic 93 % training, 7 % testing, Russian-Ukraine Conflict 90 % training, 10 % testing). We allocated 80 % of the time to training and 20 % to testing in LSTM, ARIMA and VAR. Next, we compared the average or projected price of simulations using GBM, FBM, and BB (with 1000 possible price

movements) with the real price data and the calculated performance measures.

To assess the normality of agricultural commodity price distributions, descriptive statistics such as the mean, standard deviation, and median were initially examined. A close alignment between the mean and median values typically indicates that the data may follow a normal distribution. For example, in the case of coffee prices, the mean (121,04) and median (115,65) are relatively close, suggesting an approximately symmetric distribution. These results were presented in Table 4. Similarly, for soybean and sugar, the differences between the mean and median are minimal, supporting the assumption of normality in their price distributions. In addition to central tendency measures, skewness and kurtosis values were also analyzed to evaluate the shape of each distribution. According to Tabachnick and Fidell (2013), skewness and kurtosis values between −1.5 and + 1.5 indicate acceptable levels of normality, while George and Mallery (2010) suggest a slightly broader acceptable range of ±2.0. Within this framework, commodities such as coffee, soybean, and sugar exhibit skewness and kurtosis values within or near these acceptable limits, further supporting the normality assumption. On the other hand, commodities like cocoa (skewness = 2197; kurtosis = 15,02) and cotton (skewness = 1975; kurtosis = 7592) exhibit significant departures from normality, with high levels of skewness and leptokurtic distributions, indicating the presence of extreme values or price spikes. These deviations suggest that the distributions of cocoa and cotton prices are positively skewed and have heavy tails and therefore may not conform to the assumptions of normality.

The daily returns were calculated commodity prices using the formula presented in Equation (19). In this equation, r is the return; i refers to the five commodity indexes, and P_t and P_{t-1} are adjacent prices, with P_{t-1} occurring earlier than P_t.

$$r_i = \frac{P_{i,t}}{P_{i,t-1}} \tag{19}$$

$$\mu_i = \frac{\sum r_i}{n} \tag{20}$$

$$\sigma_i = \sqrt{\frac{\sum (r_i - \overline{r}_i)^2}{n-1}} \tag{21}$$

Table 5 presents model parameter estimation. This table contains the mean return (μ) and standard deviation (σ) of the daily returns estimated using Equation (19). The formulas used for them are displayed in Equations (20) and (21), respectively. These are standard formulas typically used for estimating averages and standard deviations.

The simulated price predictions for nine periods of wheat, corn, soybean, coffee, sugar, cocoa, and cotton using three stochastic models, such as GBM, FBM, and BB, are presented in Figs. 2–15. The average simulated prices from GBM, FBM, and BB are represented by the red, blue, and yellow lines, respectively, while the actual price data is displayed in black. Fig. 2 compares the performance of GBM, FBM, and BB models in forecasting cocoa prices across nine crisis periods. GBM generally overestimates future prices due to its compounding variance, whereas FBM, with its long-memory property, tracks actual price movements more closely during moderately volatile periods. BB systematically underpredicts price spikes, reflecting its mean-reverting structure and limited responsiveness to extreme shocks including those in 2008 and 2020.

Fig. 3 compares the predictive performance of LSTM, ARIMA, VAR, GBM, FBM, and BB models for cocoa prices across nine crisis periods. LSTM closely follows sharp price changes, capturing nonlinear dynamics and structural breaks more effectively than other models. ARIMA and VAR provide smoother trajectories but tend to lag during volatile episodes. Among stochastic models GBM consistently overprojects long-term prices, FBM offers closer alignment in turbulent periods, and BB underestimates trends due to its mean-reverting nature.

Fig. 4 illustrates the simulated coffee price forecasts across nine crisis periods using GBM, FBM, and BB models. The GBM model tends to overestimate prices, particularly in volatile periods, reflecting its exponential growth assumption. FBM captures upward and downward swings more effectively, aligning better with actual trends during high-variance episodes. In contrast, BB consistently underestimates price volatility due to its mean-reverting constraint, failing to reflect sharp spikes observed during crises. Overall, FBM demonstrates greater flexibility in turbulent conditions, while GBM and BB demonstrate systematic bias.

Fig. 5 compares the forecasting performance of LSTM, ARIMA, VAR, GBM, FBM, and BB models for coffee prices across nine crisis periods. The LSTM model captures nonlinear trends and turning points relatively well, particularly during high-volatility episodes. ARIMA and VAR exhibit smoother predictions, yielding better performance in stable periods yet illustrating limited adaptability to sharp price shifts. GBM often overestimates future values, while BB underestimates volatility. FBM offers more balanced forecasts yet still trails LSTM in highly dynamic regimes. Overall, LSTM demonstrates the highest alignment with actual price movements under crisis-driven fluctuations.

Fig. 6 presents the performance of stochastic simulation models—GBM, FBM, and BB—in forecasting corn prices over nine macroeconomic periods. The GBM model yields excessively wide prediction bands, frequently resulting implausible negative prices, particularly under turbulent market conditions. FBM, while incorporating long memory, also suffers from extreme dispersion and negative forecasts, limiting its practical applicability in agricultural markets.

Fig. 7 compares the performance of LSTM, ARIMA, VAR, GBM, FBM, and BB models in forecasting corn prices across nine crisis periods. LSTM emerges as the most reliable model, capturing both trend direction and structural breaks effectively. ARIMA and VAR underperform substantially, with VAR exhibiting erratic behavior and extreme deviation from actual price levels. Among stochastic models, GBM shows a smooth upward bias, FBM oscillates with directionally inconsistent outputs, and BB fails to respond to significant price shifts.

Fig. 8 compares the forecasting performance of stochastic simulation models—GBM, FBM, and BB—for cotton prices across nine crisis periods. The GBM model exhibits extreme volatility, with simulation paths frequently diverging into unrealistic negative and overly optimistic positive values. FBM, while moderately more stable, also generates

Table 4
Normality test results.

Statistics	Cocoa	Coffee	Corn	Cotton	Soybean	Sugar	Wheat
N	10111	10111	10111	10111	10111	10111	10111
Mean	2000,925	121,040	3341	0,679	8350	12,939	4469
Std. Error	8355	0,517	0,016	0,002	0,033	0,053	0,017
Median	1930,780	115,650	2810	0,650	7430	12,040	4010
Std. Dev.	840,103	51,992	1563	0,212	3317	5360	1723
Minimum	728,650	35,500	1220	0,260	3880	3400	1920
Maximum	10566,080	328,000	8490	2110	17,900	32,570	12,790
Skewness	2197	0,8046	1226	1975	0,814	0,804	1057
Kurtosis	15,0187	0,5976	0,645	7592	−0,430	0,189	0,883

Table 5
Model parameter estimation (mean μ and volatility σ).

Periods	Days	Cocoa		Coffee		Corn		Cotton		Soybean		Sugar		Wheat	
		μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
1st	2998	0,01	0,29	0,07	0,74	0,01	0,24	0,02	0,36	0,02	0,20	0,10	0.34	0,02	0,28
2nd	3770	0,01	0,28	0,05	0,74	0,01	0,24	0,01	0,34	0,01	0,23	0,04	0,34	0,01	0,28
3rd	5595	0,01	0,29	0,04	0,61	0,02	0,25	0,01	0,33	0,01	0,22	0,03	0,35	0,02	0,30
4th	6118	0,01	0,29	0,03	0,58	0,02	0,27	0,01	0,33	0,01	0,23	0,05	0,35	0,02	0,33
5th	6900	0,01	0,30	0,05	0,56	0,03	0,28	0,02	0,33	0,03	0,24	0,06	0,35	0,03	0,33
6th	8205	0,01	0,29	0,03	0,52	0,01	0,28	0,01	0,32	0,02	0,23	0,05	0,34	0,01	0,36
7th	9010	0,01	0,28	0,02	0,50	0,01	0,28	0,01	0,32	0,01	0,23	0,04	0,33	0,02	0,35
8th	9532	0,01	0,28	0,04	0,50	0,02	0,28	0,03	0,32	0,03	0,23	0,04	0,33	0,03	0,35
9th	10110	0,04	0,27	0,04	0,49	0,01	0,28	0,01	0,32	0,02	0,23	0,04	0,32	0,02	0,35

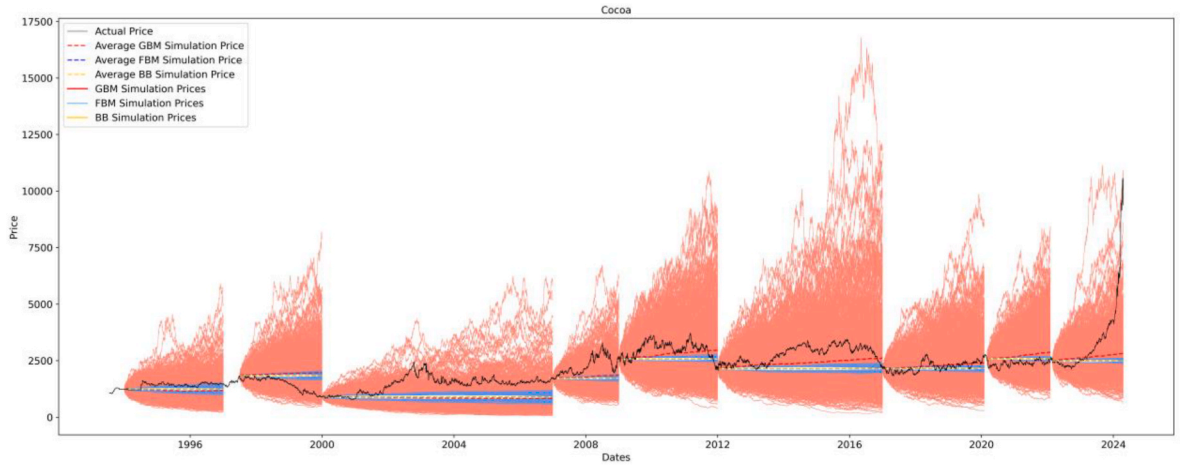


Fig. 2. Forecasting results using GBM, FBM and BB process for nine periods (cocoa).



Fig. 3. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Cocoa).

implausible forecasts under turbulent conditions. The BB model offers the narrowest prediction range, consistently underestimating actual price fluctuations due to its mean-reverting nature.

Fig. 9 compares six forecasting approaches for cotton prices across nine macroeconomic periods. Among them, LSTM demonstrates the highest predictive accuracy, closely following real price dynamics and capturing shifts during crises. In contrast, traditional models such as ARIMA and VAR perform poorly under volatility; VAR exhibits implausible predictions with significant downward bias. GBM provides stable, smoothed projections, while FBM fluctuates erratically due to its long-memory effects.

Fig. 10 illustrates the performance of GBM, FBM, and BB stochastic models in forecasting soybean prices over nine macroeconomic periods. The GBM model offers relatively balanced performance, with its mean simulations closely tracking actual price trends despite generating implausible negative values under high volatility. FBM, while incorporating long-term dependencies, displays excessive dispersion and frequent downward bias, compromising forecast realism. BB, in

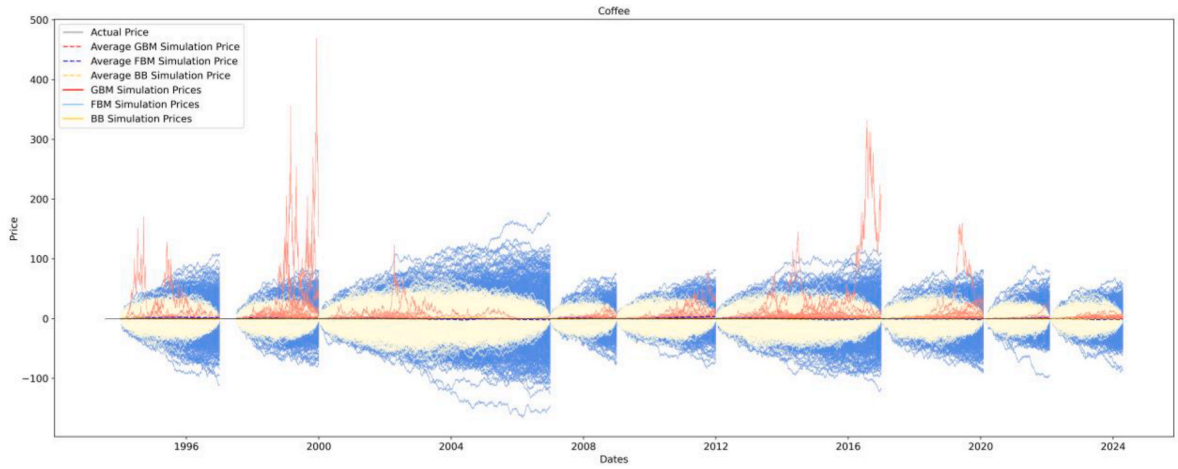


Fig. 4. Forecasting results using GBM, FBM and BB process for nine periods (Coffee).

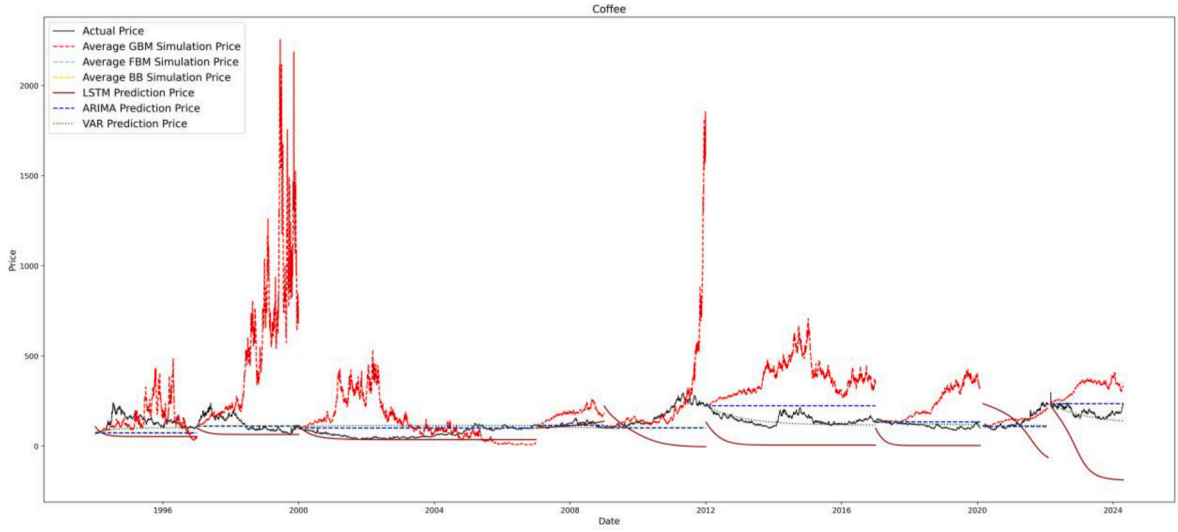


Fig. 5. Comparison of LSTM, ARIMA, VAR GBM, FBM and BB methods for nine periods (Coffee).

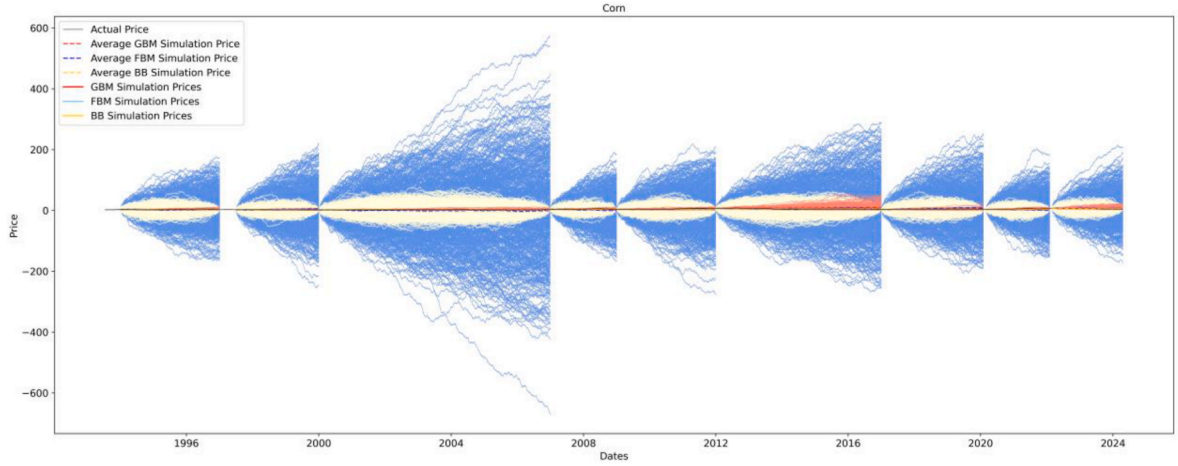


Fig. 6. Forecasting results using GBM, FBM and BB process for nine periods (Corn).

contrast, produces overly narrow simulation bands that underestimate market variability.

Fig. 11 compares the predictive performance of LSTM, ARIMA, VAR, GBM, FBM, and BB models for soybean prices across nine macroeconomic periods. The LSTM model demonstrates superior alignment with actual prices, particularly during periods of heightened volatility.

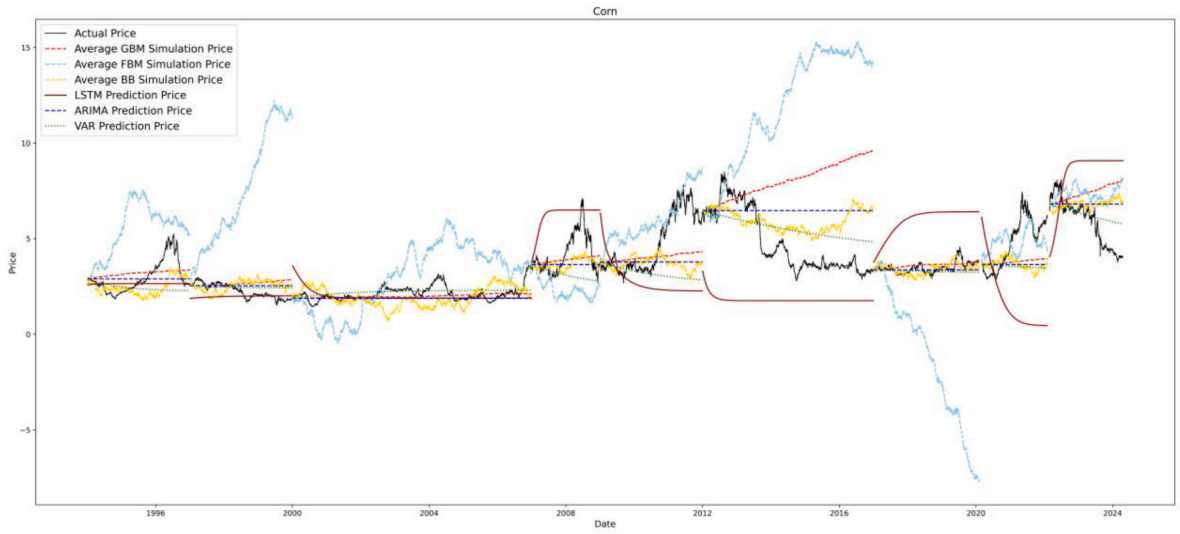


Fig. 7. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Corn).

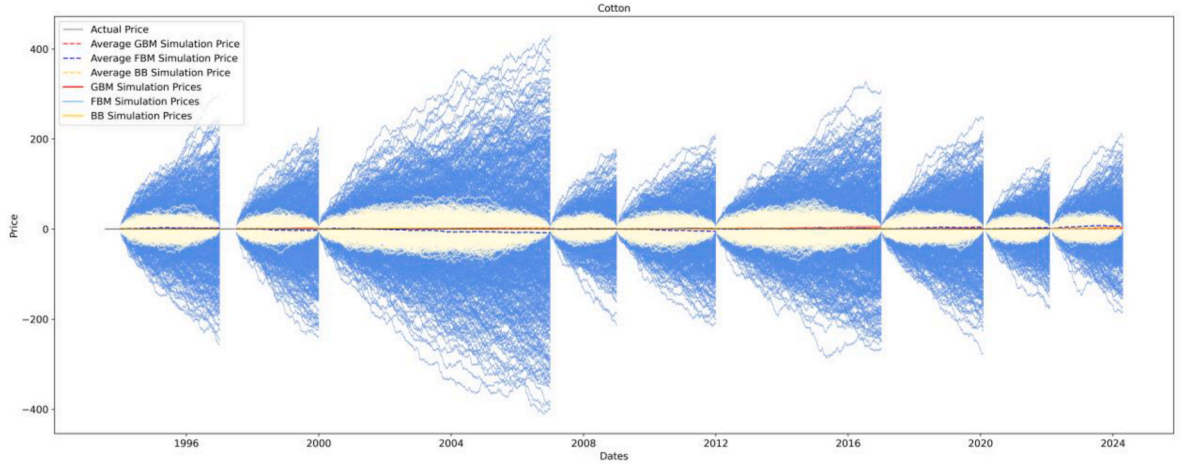


Fig. 8. Forecasting results using GBM, FBM and BB process for nine periods (Cotton).

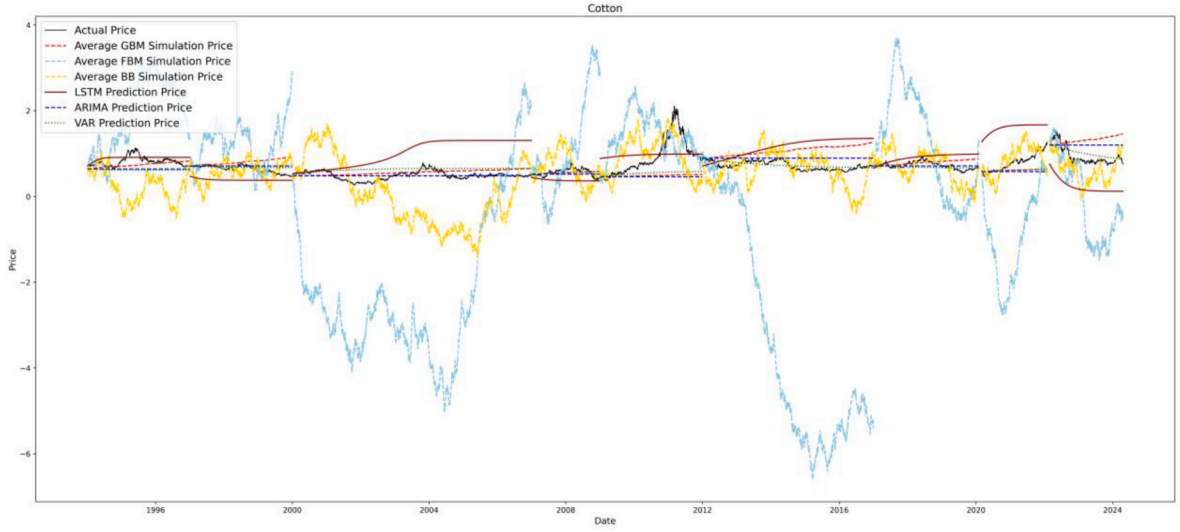


Fig. 9. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Cotton).

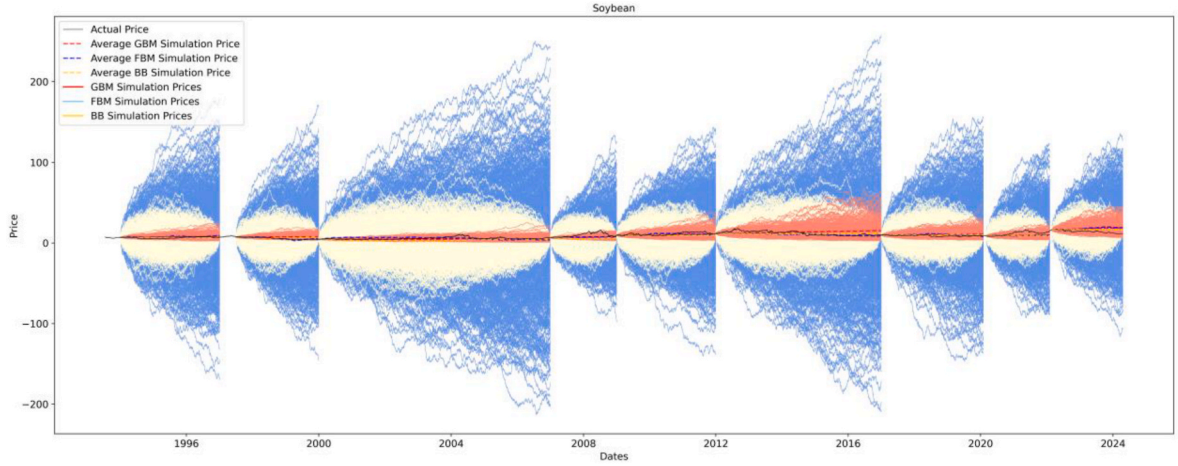


Fig. 10. Forecasting results using GBM, FBM and BB process for nine periods (Soybean).

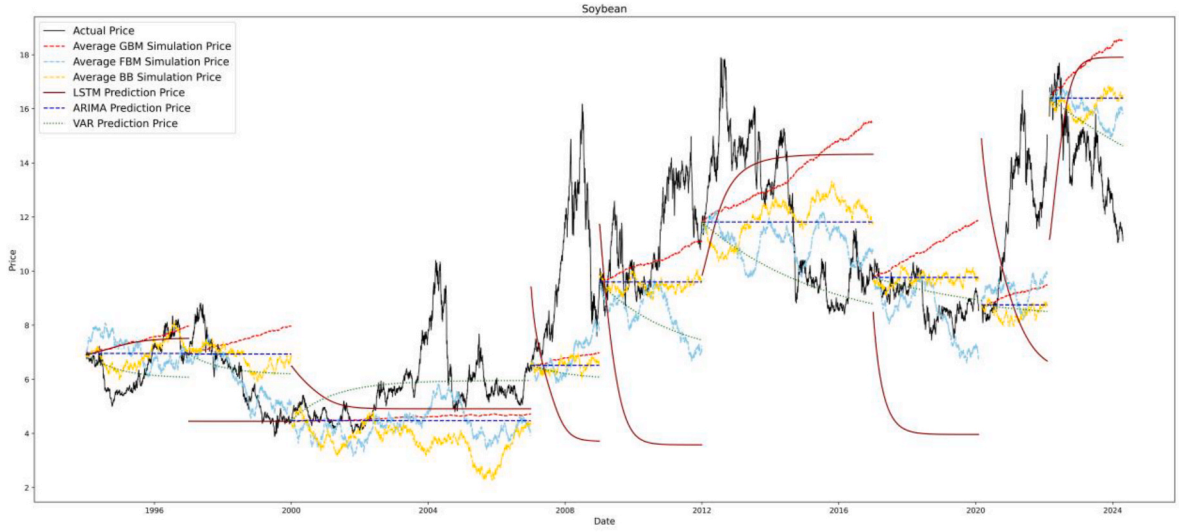


Fig. 11. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Soybean).

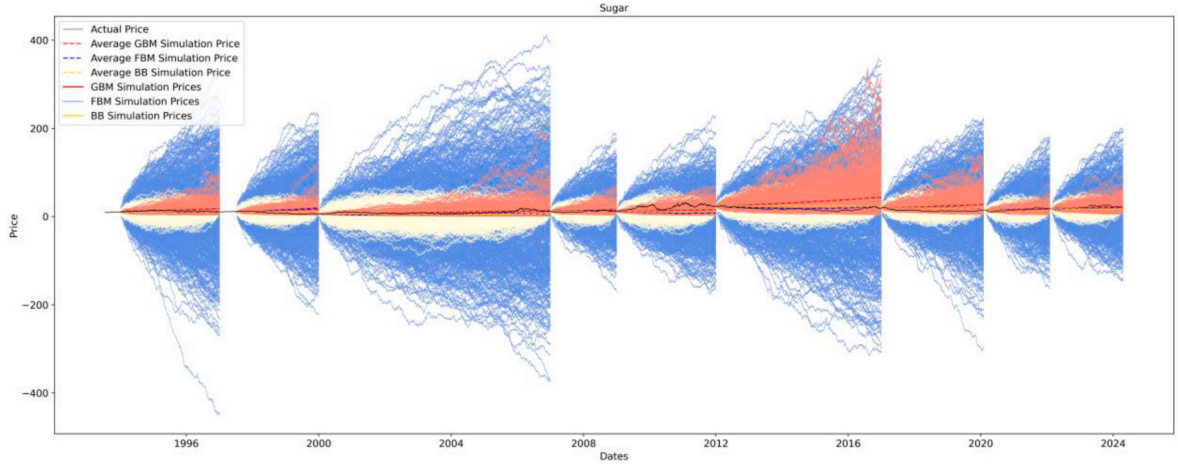


Fig. 12. Forecasting results using GBM, FBM and BB process for nine periods (Sugar).

ARIMA and VAR struggle to adapt to structural breaks, with VAR producing implausible negative forecasts in certain intervals. Among stochastic models, GBM follows upward price trends, yet tends to overpredict in calm markets, whereas FBM illustrates unstable directional accuracy. BB remains conservative throughout, failing to capture sharp price surges.

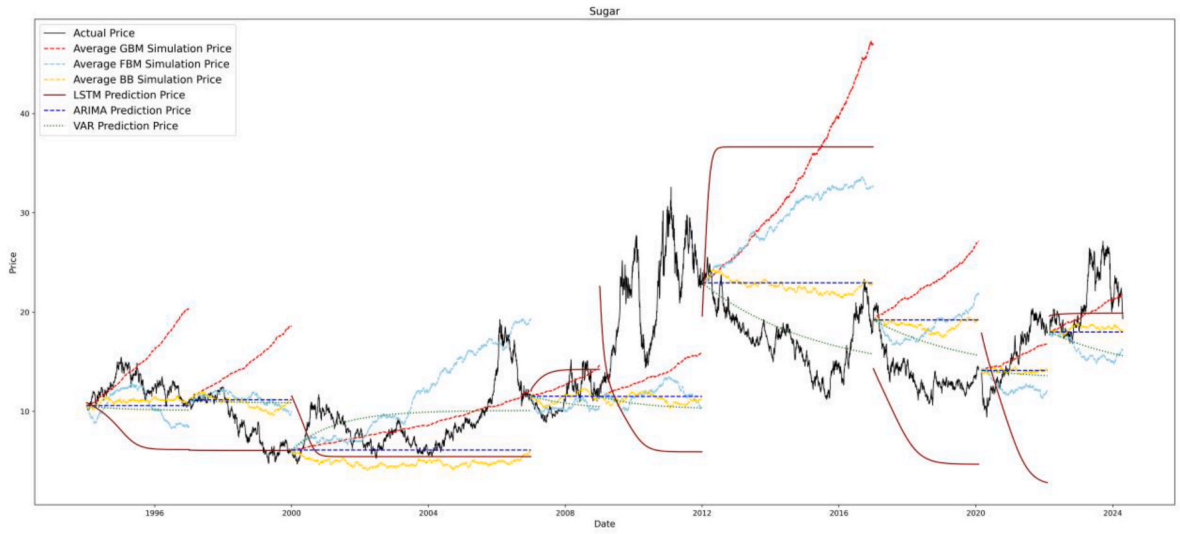


Fig. 13. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Sugar).

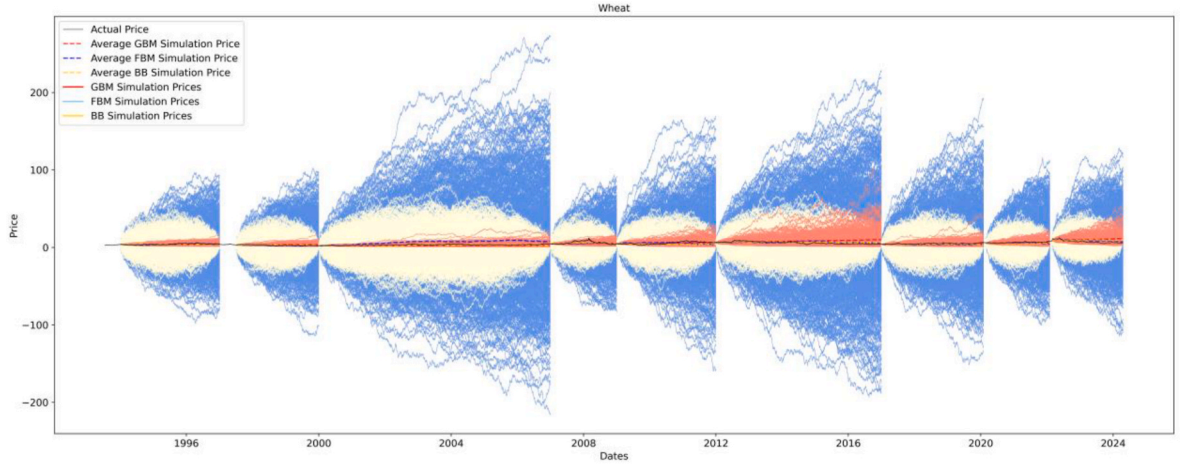


Fig. 14. Forecasting results using GBM, FBM and BB process for nine periods (Wheat).

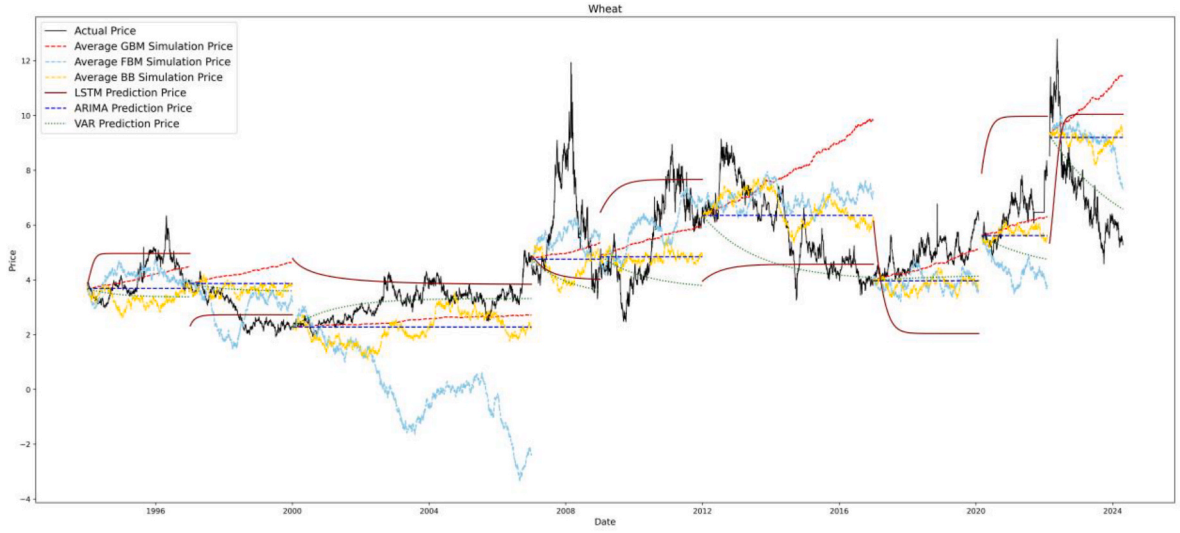


Fig. 15. Comparison of LSTM, ARIMA, VAR, GBM, FBM and BB methods for nine periods (Wheat).

Fig. 12 demonstrates the forecast performance of GBM, FBM, and BB models for sugar prices across nine macroeconomic subperiods. GBM simulations capture upward volatility well, yet tend to overestimate price levels during crises, particularly in recent years. FBM models display long-memory properties but generate implausible negative price paths, reducing their practical relevance. BB remains tightly bounded around mean values, consistently underestimating market fluctuations. GBM provides relatively realistic directional trends, while FBM and BB exhibit significant structural limitations under high-volatility conditions.

Fig. 13 compares the predictive accuracy of all models for sugar prices over nine macroeconomic periods. LSTM consistently provides closer alignment with actual price trends, particularly during volatile periods. GBM and FBM models often overshoot actual prices, especially in high-volatility episodes, while BB underrepresents market fluctuations. Traditional models such as ARIMA and VAR perform well in relatively stable environments, yet lag in turbulent phases.

Fig. 14 displays the simulation performance of GBM, FBM, and BB models in forecasting wheat prices over nine macroeconomic periods. GBM appears to best replicate actual price trends, particularly during moderate volatility phases. However, it tends to overestimate price levels under high volatility. FBM simulations exhibit the widest dispersion, reflecting the long-memory characteristics of wheat price series, yet result in inconsistent alignment with actual trends. BB simulations, while stable and narrow in range, tend to underestimate volatility and underpredict price fluctuations.

Fig. 15 illustrates the comparative forecasting performance of six models for wheat prices over nine periods. Among these, LSTM exhibits superior alignment with actual trends, capturing both the direction and magnitude of price shifts. GBM provides upward-biased yet volatility-sensitive estimates, which are beneficial in bullish markets, yet risk overestimation. FBM reflects long-memory properties but fails to capture short-term spikes. BB remains overly conservative, often underpredicting market shifts. Traditional methods such as ARIMA and VAR perform adequately during stable periods but consistently underperform in crisis contexts.

Fig. 16 illustrates a six-month-ahead wheat price simulation covering the Russia–Ukraine conflict period (2022-02-24 to 2022-08-24), using three stochastic models: GBM, FBM, and BB. The chart includes 1000 simulated trajectories for each model, visually illustrating the uncertainty range around future price paths. The black line indicates the actual observed wheat price, serving as a benchmark. For each model, the average simulated price path is overlaid (solid red for GBM, purple for FBM, and yellow for BB), while shaded areas represent the 95 % confidence intervals, capturing the spread of simulated outcomes.

Notably, the FBM model exhibits a strong upward trend and wider dispersion, reflecting its sensitivity to long-memory and volatility clustering, dynamics often amplified during geopolitical crises. In contrast, the GBM model yields a more neutral trajectory with symmetric uncertainty bands, aligning closely with the observed price early on but diverging as volatility increases. The BB model, constrained to return toward a fixed endpoint, exhibits a mean-reverting behavior that underestimates actual market shifts during the period.

This figure exemplifies the heterogeneous forecasting behavior across stochastic frameworks. While FBM captures extreme upside risk, GBM offers a stable, risk-adjusted forecast, and BB serves as a bounded scenario. The discrepancy between simulated and realized paths highlights the challenge of modeling under crisis-induced structural breaks. Out of the 756 total simulations across all models–commodity–period combinations, Fig. 16 is presented as an illustrative case to emphasize the practical implications of stochastic divergence in turbulent markets.

Fig. 17 presents a period-wise radar chart analysis of forecasting accuracy, measured by Mean Absolute Percentage Error (MAPE), for three stochastic simulation models—GBM, FBM, and BB—across seven agricultural commodities and nine global crisis periods. The corresponding numerical results are provided in Table 6, offering a detailed comparison of model performance across each period and commodity.

Across all regimes, GBM generally demonstrates superior forecasting accuracy, particularly during high-volatility events such as the COVID-19 pandemic (8th period) and the Russia–Ukraine conflict (9th period). FBM models illustrates substantial fluctuations in MAPE, with significant error spikes, especially for cotton in the 1st, 3rd, and 6th periods, indicating sensitivity to long-memory or regime-switching dynamics. The BB model, although conceptually relevant for modeling additive stochastic processes, consistently yields higher MAPE values and underperforms across nearly all periods and commodities.

Specifically, during the Commodities Boom (3rd period) and Second Global Food Crisis (6th period), FBM exhibits extreme forecast errors, most notably in cotton, where MAPE surges to several hundred percent. In contrast, GBM maintains relatively consistent and moderate error

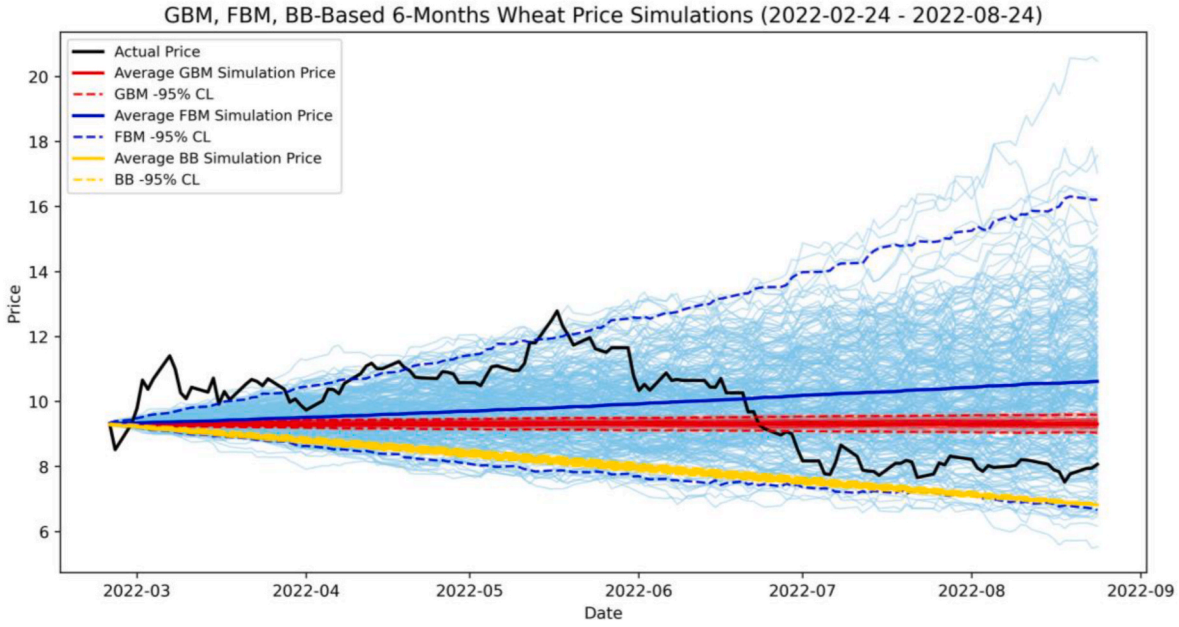


Fig. 16. Comparison of GBM, FBM, and BB methods for the Russia-Ukraine Conflict period.

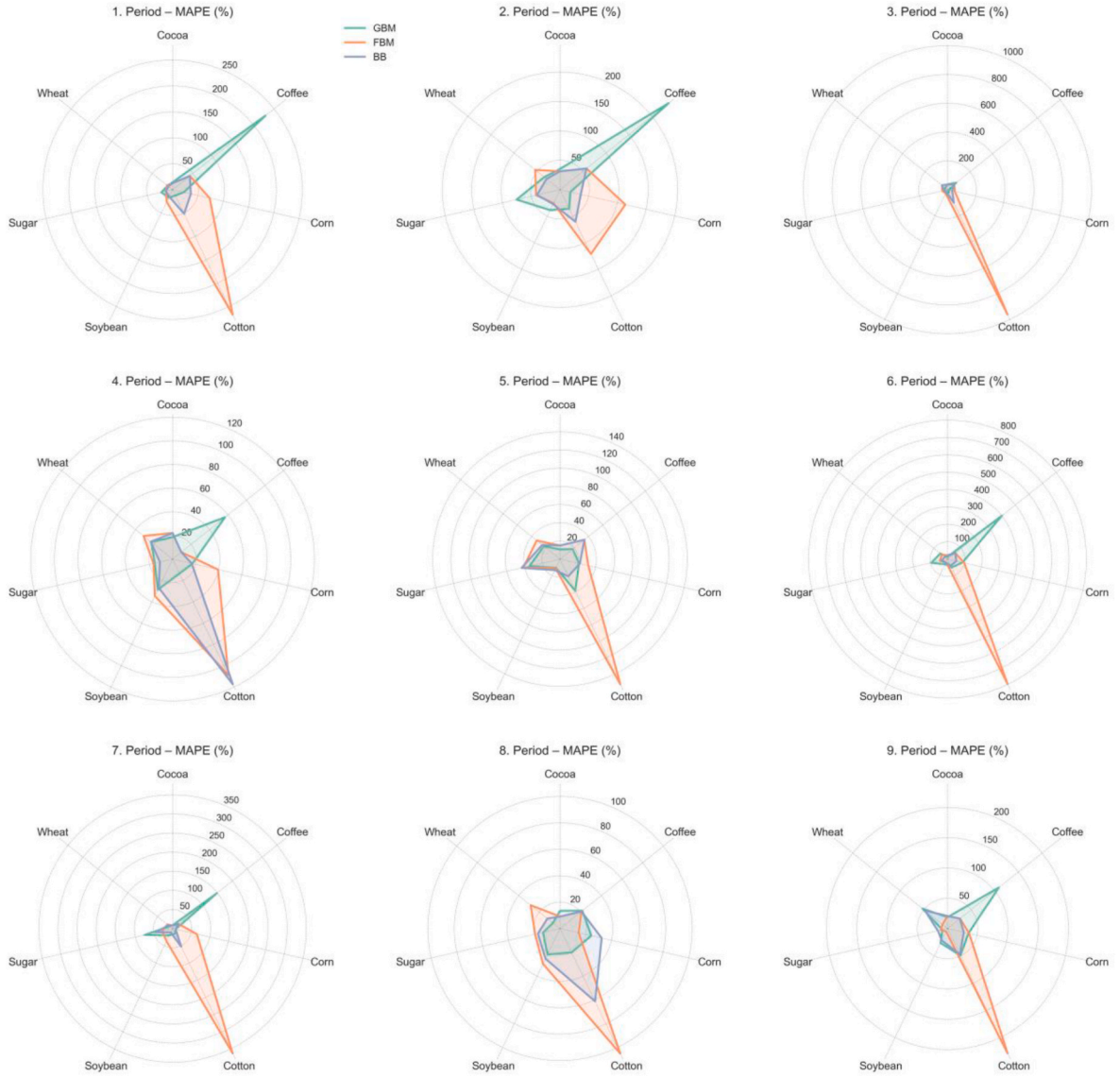


Fig. 17. Radar plots of period-wise MAPE for GBM, FBM, and BB models across agricultural commodities.

rates across all periods, underscoring its robustness under crisis-driven market structures.

The radar plots in Fig. 17 reinforce the notion that model selection should be tailored to the structural features of each regime, with GBM showing greater resilience in capturing price dynamics during systemic shocks, whereas FBM, despite its potential for modeling memory effects, may introduce instability under certain conditions. BB, as demonstrated, fails to deliver practically useable accuracy levels.

The comparative evaluation of LSTM, ARIMA, and VAR models across nine global crisis periods highlights divergent forecasting behaviors in terms of accuracy and return performance. As summarized in Table 7, LSTM often yields higher average returns in specific periods, particularly the 4th and 5th episodes for commodities like cocoa, soybean, and wheat, yet this is frequently accompanied by elevated error levels, especially during high-volatility regimes such as the 6th through 8th periods. Both MAPE and RMSE values surge significantly during these episodes, suggesting reduced predictive reliability under stress conditions. Conversely, ARIMA consistently delivers lower forecast errors across a wide range of commodities, especially during early crises (e.g., 1st and 2nd periods), though it tends to generate minimal or zero returns, implying a trade-off between predictive precision and investment profitability.

VAR displays intermediate performance in terms of accuracy but occasionally captures structural interdependence effectively, yielding relatively higher returns during the 3rd and 4th periods for commodities such as corn, cotton, and wheat. These dynamics suggest that while VAR may not consistently outperform error metrics, its ability to model inter-commodity relationships can provide a strategic advantage under regime shifts. These findings are further illustrated in Fig. 18, which presents radar plots of MAPE values across all commodities and periods. LSTM exhibits pronounced error variability, with significant spikes in volatile episodes, most notably during the 3rd and 6th periods for coffee and cotton. In contrast, ARIMA and VAR form more stable and compact distributions, particularly during mid-to-late periods such as the 5th and 7th, reinforcing their relative robustness. Overall, while LSTM's capacity to model nonlinearities is advantageous in select contexts, ARIMA's consistency and VAR's structural sensitivity offer complementary forecasting strengths across crisis regimes.

However, the analysis also highlights the challenges posed by sudden and unpredictable market changes, which significantly impact the accuracy of forecasting for all models. During such periods, error rates increase across the board, underscoring the difficulty of modeling extreme market volatility. To better evaluate the comparative performance of stochastic models (GBM, FBM, and BB) in short-term

Table 6
Comparative performance of GBM, FBM, and BB models across commodities and crisis periods.

Periods	Hurst Exp.	GBM					FBM					BB				
		MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe	MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe	MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe
Cocoa																
1st	0.678	14.138	231.360	59.175	−0.024	−0.002	12.389	203.563	9.002	0.005	0.007	12.592	206.425	0.000	0.000	0.000
2nd	0.652	35.198	530.011	47.171	0.065	0.005	31.182	470.754	4.881	0.000	0.001	31.138	470.062	0.000	0.000	0.000
3rd	0.627	37.535	704.102	73.552	−0.115	−0.003	35.024	662.969	15.272	0.002	0.002	35.029	663.227	0.000	0.000	0.001
4th	0.606	18.580	592.385	41.695	0.089	0.009	22.086	674.856	3.384	0.000	−0.001	22.065	674.399	0.000	0.000	0.000
5th	0.602	10.650	404.711	47.708	0.156	0.010	14.949	542.124	2.950	−0.001	−0.001	14.913	540.883	0.000	0.000	0.000
6th	0.611	13.360	465.556	56.291	0.289	0.011	21.562	738.383	4.982	0.001	0.001	21.593	739.482	0.000	0.000	0.000
7th	0.602	8.409	209.325	44.783	0.183	0.012	7.936	229.584	3.136	0.001	0.002	7.945	230.110	0.000	0.000	0.000
8th	0.596	13.455	331.067	36.122	0.087	0.010	9.079	239.876	1.776	0.001	0.004	9.060	239.577	0.000	0.000	0.000
9th	0.586	18.719	1703.096	42.513	0.113	0.010	20.817	1823.563	1.990	0.000	0.000	20.812	1823.418	0.000	0.000	0.000
Coffee																
1st	0.516	229.269	495.272	100.000	−0.483	0.019	42.690	72.678	55.481	−0.020	0.008	42.414	72.531	0.000	0.000	0.007
2nd	0.510	236.176	294.293	99.994	0.808	0.018	57.806	62.013	22.504	0.001	0.003	57.560	61.795	0.000	0.000	0.003
3rd	0.517	77.283	63.772	100.000	−0.548	0.016	59.667	37.582	58.128	0.001	0.007	59.355	37.374	0.000	0.000	0.005
4th	0.506	56.680	78.838	99.586	0.905	0.015	9.459	13.373	26.970	0.011	0.007	9.419	13.512	0.000	0.000	0.004
5th	0.478	17.806	48.473	99.936	1.127	0.010	34.204	92.328	29.675	0.004	0.005	34.310	92.365	0.000	0.000	0.005
6th	0.508	401.592	690.104	99.990	3.552	0.015	57.040	81.480	21.917	0.000	0.002	56.853	81.227	0.000	0.000	0.002
7th	0.511	147.884	209.341	99.658	0.362	0.012	19.642	24.257	29.322	0.003	0.005	19.968	24.607	0.000	0.000	0.004
8th	0.508	21.640	31.513	97.465	0.974	0.015	20.682	54.551	26.274	0.005	0.006	20.770	54.820	0.000	0.000	0.005
9th	0.508	108.632	223.611	97.758	1.064	0.016	26.499	52.579	12.881	0.004	0.004	26.151	51.887	0.000	0.000	0.002
Corn																
1st	0.614	22.314	0.794	41.047	0.122	0.009	73.490	2.109	2543.232	0.257	0.016	36.170	1.229	0.000	0.000	0.000
2nd	0.668	18.048	0.410	41.308	0.040	0.004	113.870	2.528	4232.954	−1.761	0.017	36.161	0.862	0.000	0.000	0.002
3rd	0.681	12.056	0.424	65.049	0.042	0.001	51.466	1.539	12023.886	0.038	0.010	33.698	0.810	0.000	0.000	0.000
4th	0.650	17.258	1.071	37.073	0.099	0.011	39.454	2.417	2104.068	−0.586	0.024	16.666	1.068	0.000	0.000	0.002
5th	0.654	21.556	1.469	42.846	0.190	0.013	32.004	2.546	2677.846	0.002	0.023	22.341	1.715	0.000	0.000	0.003
6th	0.654	88.053	3.875	48.732	0.461	0.016	98.975	4.648	1992.761	0.658	0.017	51.762	2.244	0.000	0.000	0.005
7th	0.657	8.242	0.333	43.399	0.236	0.015	65.264	2.636	3110.094	1.173	0.019	8.334	0.380	0.000	0.000	0.006
8th	0.656	24.275	1.558	36.996	0.089	0.011	14.519	0.844	2045.814	0.961	0.023	32.357	1.929	0.000	0.000	0.005
9th	2.017	35.409	2.151	37.734	0.177	0.017	36.772	2.345	1140.582	0.225	0.025	27.510	1.672	0.000	0.000	0.009
Cotton																
1st	0.672	12.589	0.150	41.913	0.311	0.017	267.234	2.659	17314.549	5.245	0.010	51.429	0.543	0.000	0.000	−0.001
2nd	0.675	35.933	0.240	41.312	0.232	0.017	121.257	0.866	15027.480	−2.642	0.016	60.607	0.425	0.000	0.000	−0.002
3rd	0.670	22.606	0.112	66.393	0.278	0.006	961.824	5.195	47387.945	−24.987	0.007	100.161	0.595	0.000	0.000	0.001
4th	0.648	13.481	0.089	43.590	0.055	0.005	108.917	0.734	14485.265	1.153	0.020	117.532	0.819	0.000	0.000	−0.001
5th	0.631	38.632	0.584	51.776	0.153	0.009	152.800	2.229	19822.591	−6.935	0.009	20.918	0.232	0.000	0.000	−0.001
6th	0.653	53.504	0.404	57.691	0.440	0.013	797.758	5.931	17685.844	−9.345	0.008	42.245	0.361	0.000	0.000	−0.001
7th	0.662	14.751	0.126	49.346	0.198	0.011	362.784	2.775	15391.616	−5.753	0.010	52.342	0.445	0.000	0.000	0.001
8th	0.655	20.097	0.241	36.352	0.138	0.014	105.130	1.164	13173.921	2.518	0.012	61.168	0.622	0.000	0.000	−0.002
9th	0.659	49.414	0.458	40.028	0.235	0.019	228.975	2.387	7489.322	−3.411	0.019	47.296	0.529	0.000	0.000	−0.005
Soybean																
1st	0.597	16.901	1.203	32.837	0.152	0.013	25.892	2.006	960.522	−0.392	0.020	14.756	1.101	0.000	0.000	0.007
2nd	0.586	38.604	2.244	29.460	0.143	0.016	25.596	1.543	764.371	−0.094	0.018	27.152	1.579	0.000	0.000	0.006
3rd	0.577	16.825	1.606	56.684	0.060	0.002	29.127	2.336	2083.086	−0.603	0.011	19.367	1.683	0.000	0.000	0.001
4th	0.566	28.728	4.158	33.146	0.068	0.009	34.509	4.611	666.083	−0.098	0.025	26.867	4.009	0.000	0.000	0.013
5th	0.580	10.525	1.633	38.471	0.167	0.013	11.259	1.701	584.736	0.001	0.019	13.406	1.791	0.000	0.000	0.012
6th	0.591	32.527	4.042	41.341	0.340	0.015	19.217	2.676	771.824	−0.163	0.016	22.588	2.825	0.000	0.000	0.007

(*continued on next page*)

Table 6 (*continued*)

Periods	Hurst Exp.	GBM					FBM					BB				
		MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe	MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe	MAPE (%)	RMSE	VaR (90 %)	Avg. Return	Avg. Sharpe
7th	0.595	20.030	1.988	38.209	0.180	0.014	36.569	3.595	753.355	−0.661	0.018	11.144	1.146	0.000	0.000	0.009
8th	0.591	21.822	3.551	32.460	0.092	0.012	29.666	4.661	651.012	−0.182	0.030	25.307	3.953	0.000	0.000	0.019
9th	0.583	26.452	4.076	31.592	0.148	0.017	5.370	0.928	349.263	−0.227	0.030	21.786	3.275	0.000	0.000	0.023
Sugar																
1st	0.690	22.260	3.659	34.262	0.850	0.034	13.144	1.871	1207.412	0.320	0.032	12.895	2.073	0.000	0.000	0.012
2nd	0.671	75.677	6.101	35.896	0.441	0.027	41.796	3.278	919.621	0.002	0.030	40.055	3.195	0.000	0.000	0.016
3rd	0.649	21.187	2.579	62.583	0.781	0.015	37.901	4.307	2726.899	0.496	0.015	24.956	3.947	0.000	0.000	0.001
4th	0.640	15.434	1.855	43.201	0.253	0.020	16.534	2.114	573.072	0.158	0.033	10.969	1.545	0.000	0.000	0.016
5th	0.635	34.489	9.183	42.435	0.420	0.021	41.357	10.683	739.439	0.225	0.026	43.315	11.347	0.000	0.000	0.013
6th	0.662	96.234	17.394	49.821	0.993	0.023	45.969	7.955	626.936	0.110	0.025	34.726	6.081	0.000	0.000	0.014
7th	0.659	74.610	10.428	46.790	0.437	0.021	28.212	3.884	544.710	−0.186	0.029	44.968	6.136	0.000	0.000	0.018
8th	0.655	13.144	2.170	42.525	0.216	0.018	19.347	3.997	505.564	−0.130	0.036	17.131	3.235	0.000	0.000	0.024
9th	0.650	8.797	2.789	43.490	0.206	0.017	11.449	3.502	412.009	0.109	0.037	16.309	4.859	0.000	0.000	0.023
Wheat																
1st	0.515	11.656	0.614	39.667	0.177	0.013	13.393	0.701	1078.533	0.024	0.017	10.388	0.488	0.000	0.000	0.003
2nd	0.546	34.188	0.946	41.926	0.137	0.011	53.976	1.510	1298.094	0.303	0.014	29.296	0.834	0.000	0.000	0.005
3rd	0.575	19.531	0.848	65.584	0.194	0.006	49.048	1.685	4041.267	1.588	0.006	49.381	2.039	0.000	0.000	0.002
4th	0.553	23.048	2.071	41.846	0.153	0.014	31.390	2.815	903.756	−0.336	0.017	23.582	2.227	0.000	0.000	0.006
5th	0.582	22.727	1.370	49.394	0.211	0.012	32.854	1.737	1178.996	0.672	0.016	25.026	1.550	0.000	0.000	0.008
6th	0.595	52.219	3.114	59.401	0.540	0.015	41.554	2.500	1398.841	0.281	0.015	20.731	1.219	0.000	0.000	0.005
7th	0.593	9.525	0.643	55.716	0.224	0.011	16.943	1.083	1721.569	0.082	0.015	13.944	0.921	0.000	0.000	0.002
8th	0.588	6.807	0.586	44.458	0.174	0.015	28.376	2.197	923.315	−0.249	0.021	12.175	1.000	0.000	0.000	0.012
9th	0.581	53.004	3.753	48.233	0.229	0.017	10.733	1.051	632.679	−0.299	0.028	47.600	3.302	0.000	0.000	0.015

1st: Farm Crisis, 2nd: Asian Financial Crisis, 3rd: 2000s Commodities Boom, 4th: Global Food Crisis, 5th: Global Financial Crisis, 6th: Second Global Food Crisis, 7th: U.S.-China Trade War, 8th: COVID-19 Pandemic, 9th: Russia-Ukraine Conflict.

Table 7
Comparative performance of LSTM, ARIMA, and VAR models across commodities and crisis periods.

Periods	LSTM			ARIMA			VAR		
	MAPE (%)	RMSE	Avg. Return	MAPE (%)	RMSE	Avg. Return	MAPE (%)	RMSE	Avg. Return
Cocoa									
1st	33.962	505.142	−0.262	12.594	206.524	0.000	12.896	211.281	−0.006
2nd	52.631	734.754	0.174	31.012	468.800	−0.001	17.947	285.947	−0.173
3rd	38.179	707.394	−0.056	35.097	664.263	0.000	20.425	412.990	0.408
4th	11.761	434.699	0.198	22.047	674.072	0.000	27.746	812.082	−0.128
5th	59.441	1926.645	−0.632	14.926	541.353	0.000	28.023	964.174	−0.283
6th	19.176	593.568	0.560	21.650	740.759	−0.001	29.333	972.086	−0.166
7th	48.340	1174.465	−0.535	7.932	229.485	0.001	10.221	295.265	−0.069
8th	33.057	793.005	0.244	9.169	241.938	0.001	6.174	186.307	−0.094
9th	34.268	1569.169	0.517	20.792	1820.295	0.003	22.880	1926.717	−0.091
Coffee									
1st	47.245	78.293	−0.097	42.118	72.078	0.000	24.506	51.911	0.384
2nd	45.918	51.189	−0.799	57.643	61.868	0.001	25.241	28.487	−0.291
3rd	91.716	56.409	0.215	60.543	38.107	0.000	79.229	49.229	0.141
4th	97.531	123.540	1.302	9.455	13.543	0.000	12.217	19.840	−0.112
5th	91.118	178.398	−1.003	34.406	92.635	0.000	33.574	91.318	0.018
6th	160.146	226.043	0.666	57.081	81.517	0.000	15.075	28.982	−0.476
7th	87.357	101.581	−0.916	19.584	24.208	0.000	10.621	14.104	−0.115
8th	84.104	112.272	1.460	20.755	54.811	0.000	18.964	50.893	0.067
9th	29.380	59.833	−0.488	26.180	51.919	0.001	10.692	26.675	−0.407
Corn									
1st	18.323	0.946	−0.134	20.717	0.834	−0.001	20.926	1.059	−0.212
2nd	23.851	0.632	−0.321	16.292	0.369	0.001	17.359	0.393	0.021
3rd	24.414	0.542	0.384	12.782	0.450	0.000	13.660	0.363	0.228
4th	19.035	1.024	0.116	17.266	1.162	0.000	23.033	1.579	−0.238
5th	47.119	2.980	−0.432	22.736	1.805	0.002	27.150	2.329	−0.247
6th	57.006	2.477	0.037	53.186	2.306	0.001	31.979	1.409	−0.253
7th	60.776	2.171	−0.628	5.809	0.297	0.000	6.764	0.347	−0.037
8th	132.960	6.474	2.393	26.182	1.712	0.000	27.474	1.812	−0.050
9th	101.959	5.673	0.698	24.083	1.511	0.000	15.603	0.998	−0.153
Cotton									
1st	10.506	0.123	0.194	20.422	0.209	0.000	23.353	0.231	−0.045
2nd	51.542	0.321	0.280	20.409	0.140	0.000	14.639	0.102	−0.079
3rd	94.198	0.440	0.948	16.769	0.098	−0.001	36.318	0.176	0.356
4th	32.582	0.216	−0.303	14.433	0.097	0.001	12.741	0.085	0.120
5th	38.739	0.594	0.083	41.986	0.621	−0.001	33.530	0.544	0.265
6th	10.905	0.112	−0.279	26.462	0.199	0.002	10.130	0.082	−0.244
7th	25.343	0.207	−0.259	9.080	0.081	0.000	9.002	0.088	−0.064
8th	30.099	0.337	−0.110	24.825	0.292	−0.003	21.566	0.259	0.090
9th	58.607	0.548	−0.760	35.940	0.330	−0.001	15.131	0.158	−0.274
Soybean									
1st	36.133	2.376	0.259	13.075	0.958	0.000	12.144	0.976	−0.127
2nd	27.461	1.595	−0.029	30.285	1.756	0.006	21.261	1.266	−0.095
3rd	38.965	2.669	−0.262	18.826	1.733	0.000	13.804	1.150	0.333
4th	19.517	3.314	0.223	30.872	4.395	0.000	33.514	4.666	−0.067
5th	33.238	3.796	0.546	14.046	2.285	0.002	24.125	3.565	−0.222
6th	44.637	5.278	0.418	19.801	2.555	0.000	16.273	2.851	−0.257
7th	40.718	3.731	−0.508	10.435	1.059	0.000	6.251	0.658	−0.085
8th	54.256	7.550	−0.489	24.091	3.901	0.000	25.007	4.042	−0.028
9th	40.232	5.756	0.209	17.589	2.713	0.001	10.861	1.723	−0.107
Sugar									
1st	20.274	2.681	0.434	14.448	2.240	0.001	17.052	2.536	−0.041
2nd	88.233	6.816	0.349	43.880	3.494	−0.001	42.137	3.353	−0.020
3rd	78.976	6.484	1.481	27.138	4.112	0.000	28.078	2.922	0.651
4th	18.119	2.957	−0.243	12.224	1.620	−0.001	13.689	1.916	−0.092
5th	25.309	7.539	0.425	43.805	11.489	0.004	47.304	12.296	−0.098
6th	120.487	20.209	0.641	35.592	6.234	0.000	13.742	2.672	−0.312
7th	89.198	12.057	0.342	42.883	5.840	0.000	26.794	3.684	−0.186
8th	98.628	15.951	1.481	17.306	3.251	0.000	18.331	3.508	−0.036
9th	38.892	9.601	−0.381	14.508	4.393	0.000	20.184	5.768	−0.133
Wheat									
1st	12.903	0.717	0.052	13.540	0.806	0.000	16.933	0.988	−0.084
2nd	95.726	2.570	0.677	26.053	0.712	0.001	33.683	0.928	0.099
3rd	15.116	0.582	0.406	26.427	1.089	0.000	10.768	0.464	0.456
4th	23.709	2.200	0.022	24.046	2.261	0.001	30.493	2.827	−0.236
5th	105.220	5.240	1.255	25.569	1.696	−0.007	28.559	2.236	−0.222

(*continued on next page*)

Table 7 (*continued*)

Periods	LSTM			ARIMA			VAR		
	MAPE (%)	RMSE	Avg. Return	MAPE (%)	RMSE	Avg. Return	MAPE (%)	RMSE	Avg. Return
6th	56.562	3.741	−0.652	24.758	1.495	0.006	18.529	1.567	−0.353
7th	43.587	2.323	−0.370	17.723	1.071	−0.004	15.514	0.960	0.039
8th	66.879	4.269	0.946	12.448	1.007	0.005	18.942	1.530	−0.149
9th	41.129	2.873	0.030	36.354	2.562	−0.012	13.921	1.154	−0.293

1st: Farm Crisis, 2nd: Asian Financial Crisis, 3rd: 2000s Commodities Boom, 4th: Global Food Crisis, 5th: Global Financial Crisis, 6th: Second Global Food Crisis, 7th: U.S.-China Trade War, 8th: COVID-19 Pandemic, 9th: Russia-Ukraine Conflict.

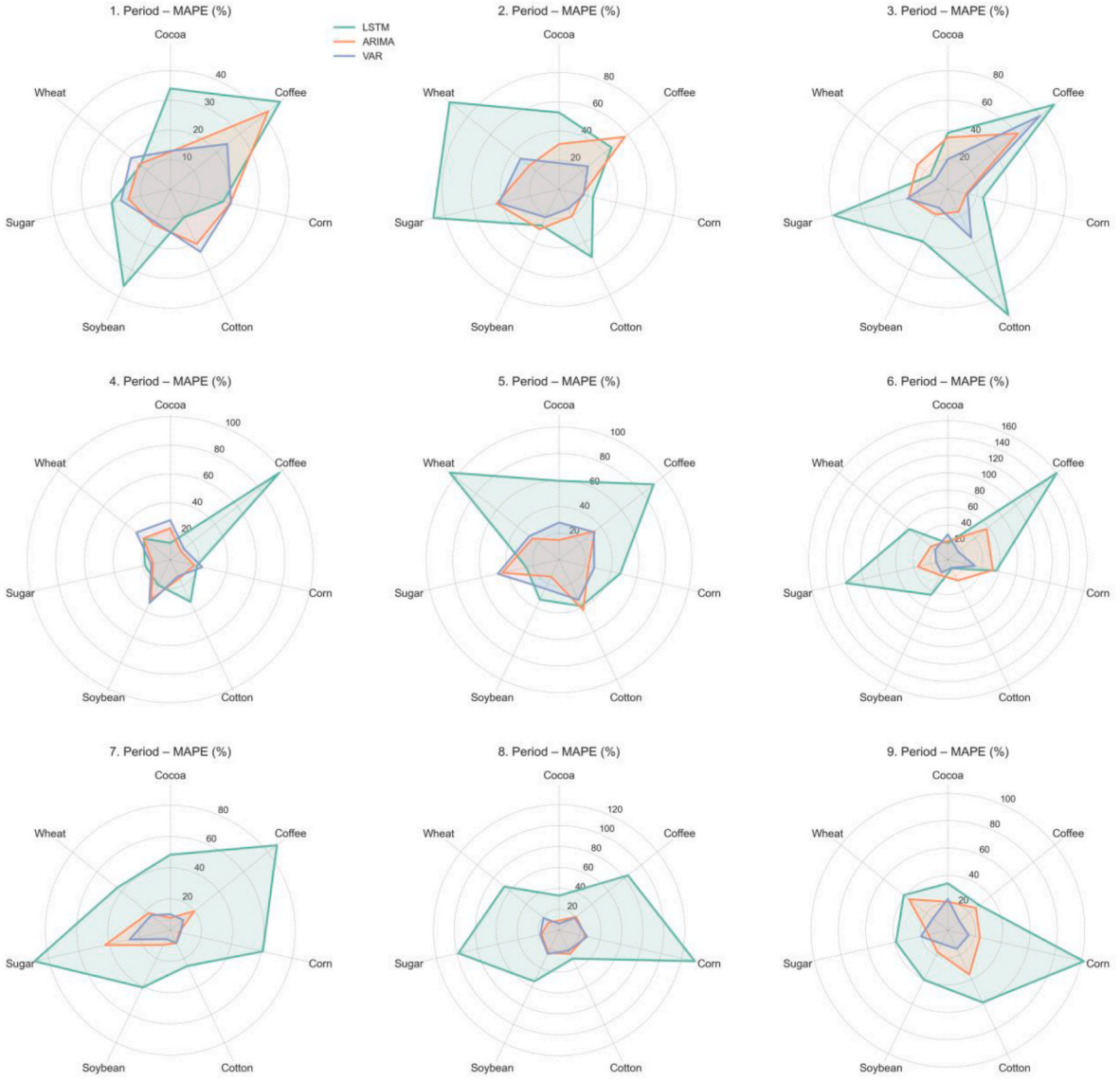


Fig. 18. Radar plots of period-wise MAPE for LSTM, ARIMA, and VAR models across agricultural commodities.

predictions, we focused on the period of the Russia–Ukraine conflict—a macroeconomic period marked by high uncertainty and significant market disruptions. This approach enables a more in-depth examination of model resilience under volatile conditions.

During the Russia-Ukraine Conflict period, the forecasting performance of GBM, FBM, and BB is evaluated for various energy commodities. The analysis spanned short-to medium-term horizons, from one month to six months, to evaluate how predictability changes over time. By dividing the period into subintervals, the study aimed to capture the dynamic nature of the energy markets and their response to stochastic processes. Given the randomness embedded in the stochastic differential equations used by these models, a single simulation path is insufficient for robust validation. Instead, multiple simulation runs (N = 1, 10, 100, and 1000) were conducted for each model, and the MAPE was calculated and averaged across all possible paths.

Table 8 presents the performance of GBM, FBM, and BB in forecasting the prices of cocoa, coffee, corn, cotton, soybean, sugar, and wheat over different time horizons. The results show that, across 1000 simulation runs, the average MAPE for all three models remained below 10 % for both the one-month and six-month horizons. This indicates that

Table 8
MAPE values for the Russia-Ukraine Conflict period (1–6 months).

Commodities	Simulation Runs	1 month			6 months		
		GBM	FBM	BB	GBM	FBM	BB
Cocoa	1	10.923	1.808	1.792	8.858	5.564	5.905
	10	1.457	1.812	1.801	8.575	5.795	5.902
	100	2.031	1.817	1.803	5.668	5.923	5.902
	1000	1.951	1.805	1.802	7.218	5.908	5.898
Coffee	1	27.123	5.829	4.979	47.993	4.653	6.457
	10	6.008	6.286	5.779	27.478	6.145	6.215
	100	5.674	6.050	6.007	18.331	6.176	6.363
	1000	7.791	5.908	5.966	9.915	6.553	6.425
Corn	1	6.254	93.814	11.732	10.845	376.549	74.601
	10	5.004	16.545	5.174	11.702	70.912	11.558
	100	5.973	13.925	5.141	8.278	21.285	10.016
	1000	5.797	4.516	5.734	9.121	9.872	9.071
Cotton	1	5.347	98.058	99.702	14.614	985.503	401.401
	10	3.506	94.851	32.441	11.519	665.431	100.018
	100	2.775	66.386	14.493	10.628	42.066	47.244
	1000	2.792	4.507	5.090	9.336	9.962	8.843
Soybean	1	1.612	21.035	5.434	8.059	64.282	30.198
	10	3.109	4.200	2.305	5.134	12.707	8.342
	100	1.090	1.135	1.195	4.800	10.262	5.942
	1000	1.052	1.291	1.170	4.443	4.432	4.271
Sugar	1	2.955	7.177	5.879	12.929	26.175	12.111
	10	2.186	3.785	4.184	4.714	73.928	8.426
	100	3.774	6.398	5.083	4.507	8.136	8.268
	1000	4.213	3.761	4.419	4.832	5.461	5.317
Wheat	1	9.033	19.850	12.991	19.628	78.697	29.473
	10	9.910	6.130	14.229	15.934	24.962	17.012
	100	9.177	11.313	11.664	15.106	20.524	13.676
	1000	9.005	9.103	9.963	9.355	9.522	9.272

Table 9
MAPE values for the Russia-Ukraine Conflict period (1–6 months).

Commodities	1 month			6 months		
	LSTM	ARIMA	VAR	LSTM	ARIMA	VAR
Cocoa	7.779	1.961	1.840	40.139	6.130	4.762
Coffee	15.578	6.014	4.244	5.344	6.504	5.419
Corn	2.961	6.029	6.363	38.576	8.193	8.887
Cotton	6.685	2.916	3.383	48.980	10.385	11.024
Soybean	10.710	1.144	1.310	4.448	4.211	3.965
Sugar	10.899	4.734	5.065	24.002	5.222	6.625
Wheat	13.863	11.261	11.349	14.214	13.837	12.770

the models demonstrated robust forecasting performance during the selected crisis period.

Table 9 presents MAPE values for LSTM, ARIMA, and VAR models during the Russia–Ukraine Conflict over 1 and 6-month horizons. ARIMA and VAR outperform LSTM in short-term forecasting across most commodities, especially cocoa, soybean, and cotton. Overall, ARIMA offers stable short-term precision, while VAR exhibits balanced performance across both horizons. LSTM's performance remains sensitive to commodity type and forecasting horizon.

Fig. 19 illustrates the comparative forecasting accuracy of stochastic (GBM, FBM, BB) and statistical ML (LSTM, ARIMA, VAR) models across seven agricultural commodities over one-month and six-month horizons during the Russia–Ukraine Conflict.

In the top panels (GBM, FBM, BB), all three models exhibit relatively similar short-term (one-month) MAPE performance, with BB and FBM outperforming GBM for cocoa, coffee, and soybeans. Over the six-month horizon, FBM illustrates increased error rates for coffee and corn, while BB remains more stable and generally exhibits the lowest MAPE across most commodities.

In the bottom panels (LSTM, ARIMA, VAR), the LSTM demonstrates substantially higher MAPE values, particularly for cocoa, cotton, and corn, across both time horizons, indicating reduced forecasting accuracy under volatile conditions. ARIMA and VAR models consistently deliver lower MAPE values, with VAR illustrating marginal improvements in wheat and sugar forecasts. Overall, ARIMA appears to deliver the most stable and accurate forecasts during geopolitical disruptions, while LSTM's performance deteriorates in medium-term projections.

5. Conclusion and policy implications

This study provides a comprehensive comparative assessment of stochastic simulation models (GBM, FBM, BB) and statistical/machine learning approaches (LSTM, ARIMA, VAR) in forecasting agricultural commodity prices across nine major global crisis periods, including the most recent Russia–Ukraine conflict. Using performance metrics such as MAPE, RMSE, VaR, average returns, and Sharpe ratios, the analysis highlights model-specific advantages and limitations under diverse macroeconomic shocks and market regimes.

Emphasized GBM consistently for stronger phrasing outperforms other stochastic models in terms of forecast accuracy and risk-adjusted returns, particularly during turbulent episodes such as the Global Food Crisis and the Russia–Ukraine conflict. A representative simulation from that conflict period (Fig. 16) supports these findings: GBM closely tracked actual wheat price trends, while FBM exhibited wider uncertainty due to long-memory effects and BB failed to capture directional shifts. This underscores GBM's relative robustness under crisis conditions and illustrates that forecasting performance is highly period dependent.

Among data-driven approaches, ARIMA demonstrates robust and stable predictive accuracy across short- and long-term horizons, particularly during early-stage or moderate crises. LSTM, while capable of modeling complex dynamics, demonstrates substantial error inflation under extreme volatility, such as in the 6th–8th periods. This aligns with existing literature demonstrating the superior accuracy of deep learning in financial time-series forecasting. For instance, Kamdem et al. (2020)

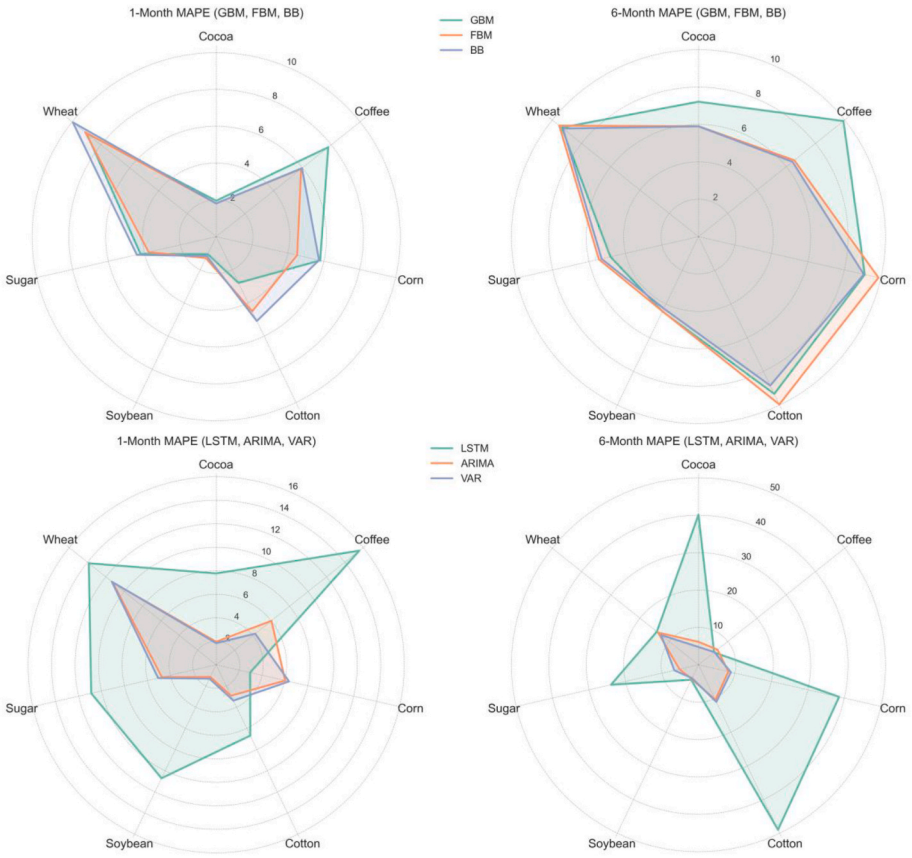


Fig. 19. Radar plots of MAPE values across models and commodities.

report that an LSTM can forecast commodity prices with very high accuracy (e.g. ~98 % for crude oil), and Ameur et al. (2024) similarly find LSTM to be an effective tool for commodity index prediction. Meanwhile, VAR models perform moderately, with relative strength in capturing interdependencies during structurally complex events like the Commodities Boom or U.S.–China Trade War.

Our findings also align with the prevailing literature by highlighting situations where traditional approaches retain value. Amid crisis episodes characterized by abrupt structural breaks, simpler models such as ARIMA or even a random-walk baseline at very short horizons were sometimes competitive with, or even briefly outperformed, complex algorithms. This outcome is consistent with recent observations that over relatively brief forecast horizons or immediately after a shock, parsimonious models can adapt faster to regime changes that confound over-parameterized learners (Vancsura et al., 2023). For instance, Skehin et al. (2018) demonstrated that a basic ARIMA could outperform a wavelet-enhanced LSTM for next-day stock price forecasts under certain conditions. In line with this, our multi-crisis analysis reveals that no model is universally resilient, and forecast accuracy declined significantly during high-volatility episodes such as the Global Food Crisis, the 2008 Financial Crisis, and the COVID-19 outbreak. This corroborates findings by Vancsura et al. (2023), who reported a substantial increase in MAPE during crisis years compared to stable periods. Nonetheless, the relative advantage of LSTM and other AI-based models remained evident, as they often achieved the lowest forecasting errors even under extreme conditions. This reinforces prior research, such as Siami-Namini et al. (2018, pp. 1394–1401), affirming the superior adaptability of LSTM models in dynamic market environments, albeit with reduced margins during episodes of systemic disruption.

Beyond raw accuracy, our study offers new insights on return predictability and market behavior across different commodity groups and crisis contexts. Consistent with Amin et al. (2024), we observe heterogeneity in how commodities react to crises. In our analysis, staple grains tended to exhibit sharper price swings and more frequent structural breaks during global shocks compared to commodities like livestock or tropical beverages. This aligns with Amin et al.'s finding of greater volatility and structural breaks in plant-based commodities versus animal-based during the 2008 recession and COVID-19 pandemic.

With regard to agricultural policies, these findings offer both cautionary and constructive implications for enhancing agricultural market stability and food security. First, the clear drop in forecast accuracy during crises indicates that policymakers must avoid reliance on any single model or point forecast as a precise predictor in volatile conditions. Unprecedented shocks are typically associated with heightened levels of market uncertainty. Thus, prudent risk management for food security calls for building policy buffers and response strategies that anticipate a range of possible price outcomes. For example, given that even the best models saw errors rise to several percentage points MAPE in crises, governments should consider implementing protective measures like strategic grain reserves or flexible import tariffs to mitigate the impact of forecast errors on vulnerable populations. Similarly, our results affirm that integrating advanced analytics can substantially improve early warning capabilities. The superior performance of LSTM and related AI models indicates that agencies should invest in modern forecasting tools and expertise. By deploying state-of-the-art models, authorities can gain more timely insights into emerging price trends, for instance, an LSTM-based early warning system could potentially detect an upcoming price surge faster than legacy methods, buying time to intervene. In addition, the adoption of hybrid and ensemble forecasting approaches is highly recommended. Our study corroborates others in finding that hybrids can leverage the complementary strengths of different models. Ray et al. (2023) demonstrate that an ARIMA–LSTM hybrid reduced error by over 30–70 % compared to single models on

volatile agricultural price series. Policymakers could promote the development of such ensemble models in commodity monitoring, for instance, combining econometric models with neural networks to improve robustness. The goal should be to create forecasting systems that are resilient to structural breaks and regime changes, perhaps by continuously retraining on the latest data and incorporating event-specific predictors when relevant.

Crucially, the differentiated commodity behavior observed means that tailored policy responses. Commodities such as staple grains, which our study and prior research flag as highly responsive and volatile in crises, warrant special attention. Policymakers might prioritize transparent stockholding and import coordination for grains, since these markets can experience sudden supply-demand imbalances. Meanwhile, commodities exhibited more stability or lagged reactions may benefit more from producer-focused support aimed at maintaining long-term supply chain viability. Additionally, improved information sharing and market transparency mitigate price volatility. Since exogenous news (e.g. trade policy announcements, conflict developments) exert considerable influence on prices, creating platforms for the timely dissemination of such information will enhance both model-based forecasts and market participants' expectations, reducing panic-driven volatility.

Our results both corroborate and extend the existing literature by demonstrating that advanced forecasting models, particularly LSTM and hybrid approaches, yield substantial improvements in accuracy and predictive reliability across a range of global crises. However, the persistence of forecasting challenges under extreme volatility highlight that no single model is universally robust. This highlights the need for policymakers to adopt a diversified, context-sensitive forecasting framework that balances complexity, interpretability, and adaptability.

Given the pronounced influence of external shocks on agricultural markets, future research should advance toward multivariate forecasting frameworks that incorporate exogenous variables such as climate anomalies, supply chain disruptions, and policy interventions. Integrating these drivers into adaptive models may enhance forecast timeliness and precision, particularly in crisis-prone commodity sectors including staple grains, thereby supporting more informed and resilient policy responses.

Since no single forecasting model consistently outperforms across all crisis regimes, a flexible, context-sensitive modeling framework constitutes a fundamental component of agricultural price monitoring and early warning systems. During structurally stable periods or short-term fluctuations, traditional models such as ARIMA may suffice due to their simplicity and robustness. In contrast, under conditions of heightened uncertainty, such as pandemics, trade disruptions, or geopolitical shocks, more sophisticated models such as LSTM or FBM models warrant greater emphasis to better capture nonlinear dynamics and structural shifts.

Public institutions and commodity boards are encouraged to invest in adaptive forecasting infrastructures capable of integrating multiple models and dynamically adjusting to crisis typologies based on early diagnostic indicators. Such systems can substantially improve the responsiveness, timeliness, and accuracy of policy interventions aimed at food price stabilization and market resilience. By embedding AI-driven forecasts into early warning platforms, tailoring stabilization policies to specific commodities, and establishing contingency measures for forecast deviations, authorities can better manage the volatility of global food price cycles. In an era characterized by overlapping crises, from financial instability to pandemics and geopolitical conflicts, strengthening analytical capacity while maintaining prudent policy buffers is indispensable for ensuring agricultural market stability and food security. The convergence of our findings with existing empirical research offers a strong basis for these forward-looking policy frameworks and highlights the importance of combining technical innovation with institutional readiness.

Future extensions of this study should explore multivariate forecasting models that integrate price series with climate variables (e.g., precipitation, temperature anomalies), macroeconomic indicators (e.g., inflation, trade volume), and event-based data (e.g., conflict indices, pandemic metrics). Leveraging machine learning feature selection methods can contribute to identifying the most relevant predictors and further improve model generalizability across different crisis regimes.

Declarations

Conflict of interest: The authors have no confict of interest to disclose regarding the content of this article.

Author contributions

Conceptualization, B.A.C. and S.C.; methodology, S.C.; software, S.C.; formal analysis, S.C. and B.A.C.; investigation, B.A.C.; data curation, S.C.; writing—original draft preparation, S.C. and B.A.C.; writing—review and editing, B.A.C.; visualization, S.C.; supervision, B.A.C.

All authors have read and agreed to the published version of the manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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